LaTeX, an introduction

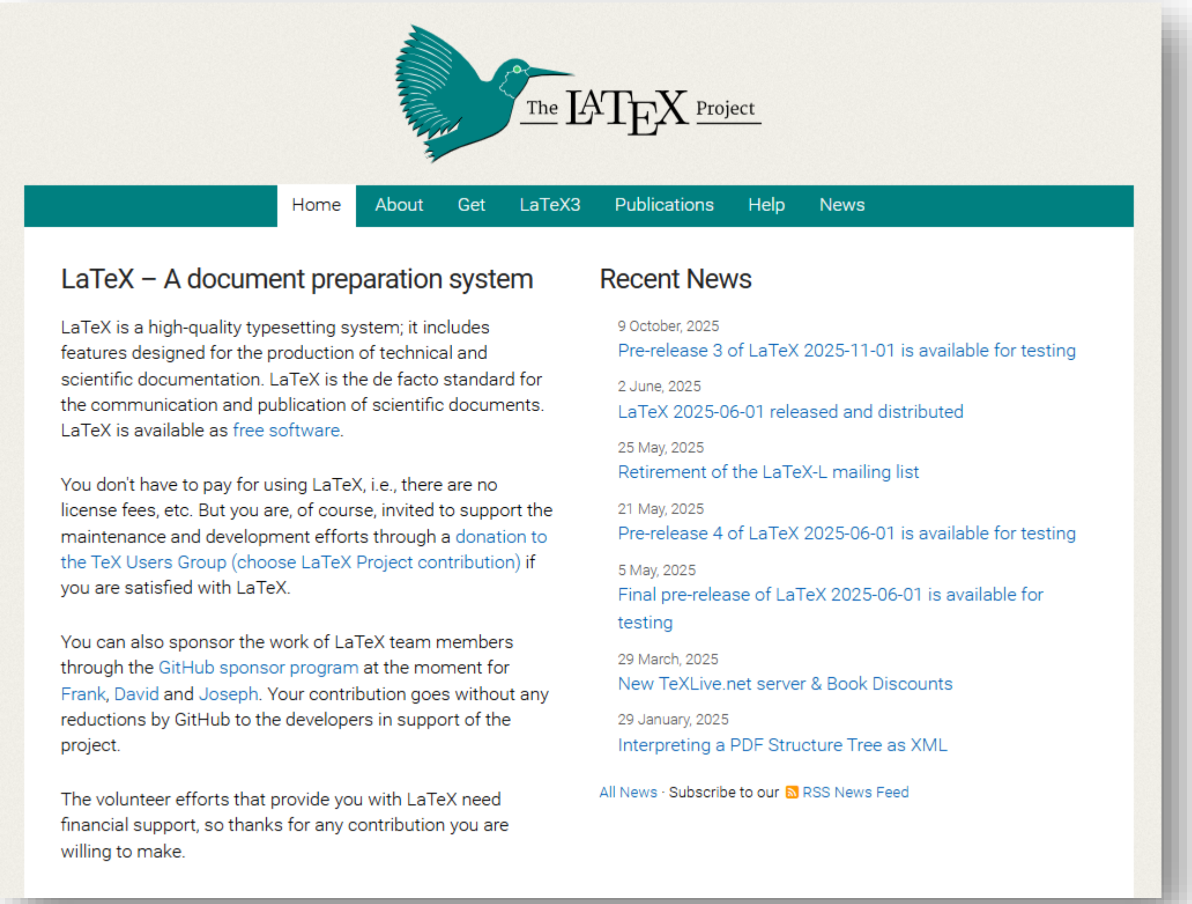
Jérôme LANDRÉ

Summary

1. What is LaTeX?
2. Why LaTeX?
3. How to use LaTeX?
4. LaTeX first article
5. Several LaTeX commands
6. LuaLaTeX
7. Bibliography
8. Using a template
9. PGF/TikZ

1. What is LaTeX?

- LaTeX is a powerful **typesetting system** designed for creating **professional**, **high-quality** documents with **precision** and **consistency**. Unlike traditional word processors, LaTeX separates content from formatting, giving you complete control over document structure and appearance.
- It's the gold **standard** in **academia**, **science**, **engineering**, and technical fields for producing complex documents with mathematical equations, cross-references, and bibliographies.



Source: https://www.latex-project.org

1. What is LaTeX?

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→ **Origins of TeX**

- 1968–1976: **Donald Knuth** publishes the early volumes of *The Art of Computer Programming*. The first editions use traditional metal typesetting, but later volumes switch to phototypesetting, which Knuth found inferior in quality.
- March 30, 1977: Knuth receives disappointing proofs of his next book volume and decides to design his own digital typesetting system.
- May 13, 1977: Knuth drafts the first design notes describing TeX — a system capable of high-quality mathematical typesetting.
- 1978: The first working version, called TeX78, is developed at Stanford University in the SAIL programming language.
- 1979: Knuth releases TeX and its companion font system METAFONT, along with a manual, under the American Mathematical Society's publication series.

→ **Maturation and Standardization**

- 1982–1989: TeX is rewritten in WEB, Knuth's literate programming system, producing TeX82, the stable version used today. It introduces better error handling and exact reproducibility of results across platforms.
- 1990: Knuth freezes TeX's core code and guarantees that any future versions will keep identical output, marking TeX as a "completed system." Its version number now asymptotically approaches π (3.141592...).

Sources: Perplexity AI, https://www-cs-faculty.stanford.edu/~knuth/

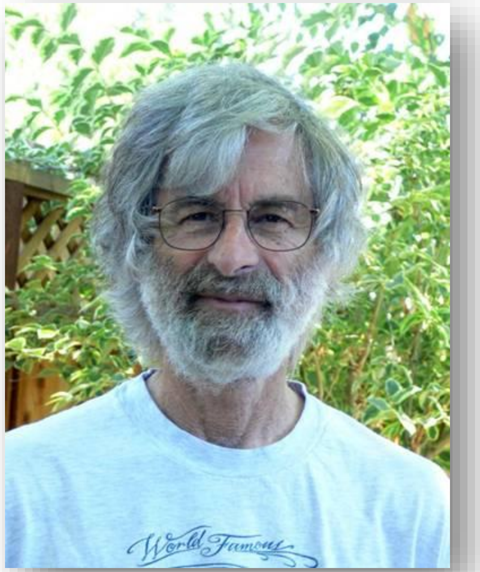
1. What is LaTeX?

→ **The Birth of LaTeX**

- 1983: **Leslie Lamport**, recognizing TeX's complexity for users, develops LaTeX, a macro-based layer that provides structured document commands (sections, lists, references) while using TeX for typesetting.
- 1985: The first official edition of *LaTeX: A Document Preparation System* is published, making LaTeX the de facto standard for academic typesetting worldwide.
- 1994: The LaTeX2ε version unifies earlier variants of LaTeX and becomes the maintained standard still used today.

→ **TeX and LaTeX in Modern Use**

- 2008: LuaLaTeX integrates the scripting power of Lua, enabling dynamic programming within documents.
- 2025: TeX remains in active use globally, with the latest TeX Live distribution updated in March 2025.



Source: Perplexity AI, https://www.lamport.org/

2. Why use LaTeX?

Mathematical Excellence
Superior handling of complex mathematical equations, symbols, and formulas with pixel-perfect rendering

Automated Formatting
Automatic numbering, cross-referencing, table of contents, and bibliography generation

Professional Quality
Consistent, publication-ready output with precise control over typography and layout

Focus on Content
Write without worrying about formatting—LateX handles the visual design automatically

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2. Why use LaTeX?

- **Pros**
  - LaTeX is **free** software
  - Very **professional** result
  - Less management for your documents
  - A **text editor** is sufficient
  - **Independent** from the computer OS
  - Elements are placed **automatically**

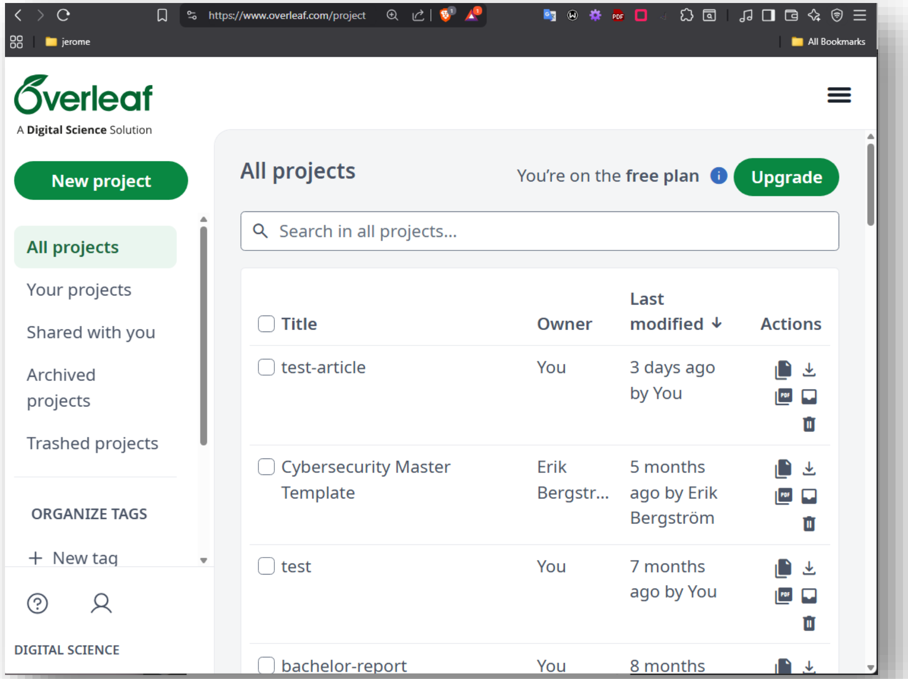
- **Cons**
  - **Not** WYSIWYG (What You See Is What You Get)
  - A new **language** to learn
  - **Debugging** problems can be tricky
  - Elements are placed **automatically**

Source: https://nitens.org/w/latex/, https://www.andy-roberts.net/latex/benefits/

3. How to use LaTeX?

- **Locally**
  - LaTeX is **free** software
  - You can **download** and **install** a distribution on your local computer (Windows, MacOS, Linux…): TexLive for instance
  - Visual Studio Code with LaTeX extensions
  - TexStudio, TexMaker, TexWorks, TeXpert…

- **Online**
  - **overleaf**



Source: https://www.tug.org/texlive/, https://get.alternative.to/category/education-reference/latex

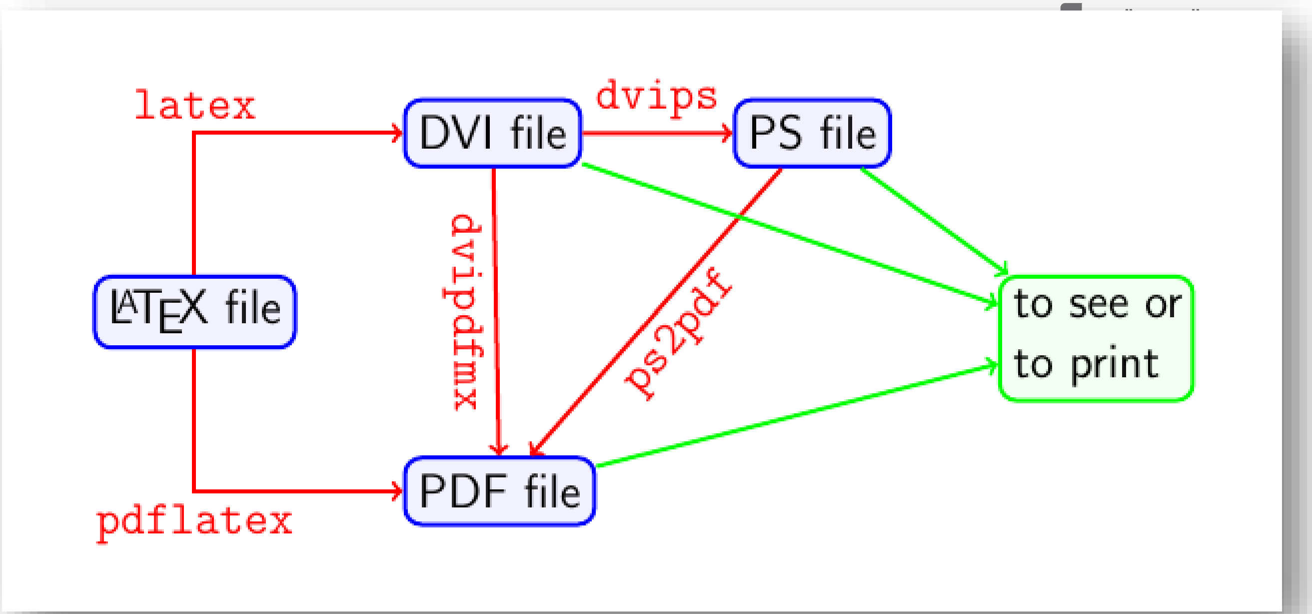
3. How to use LaTeX?

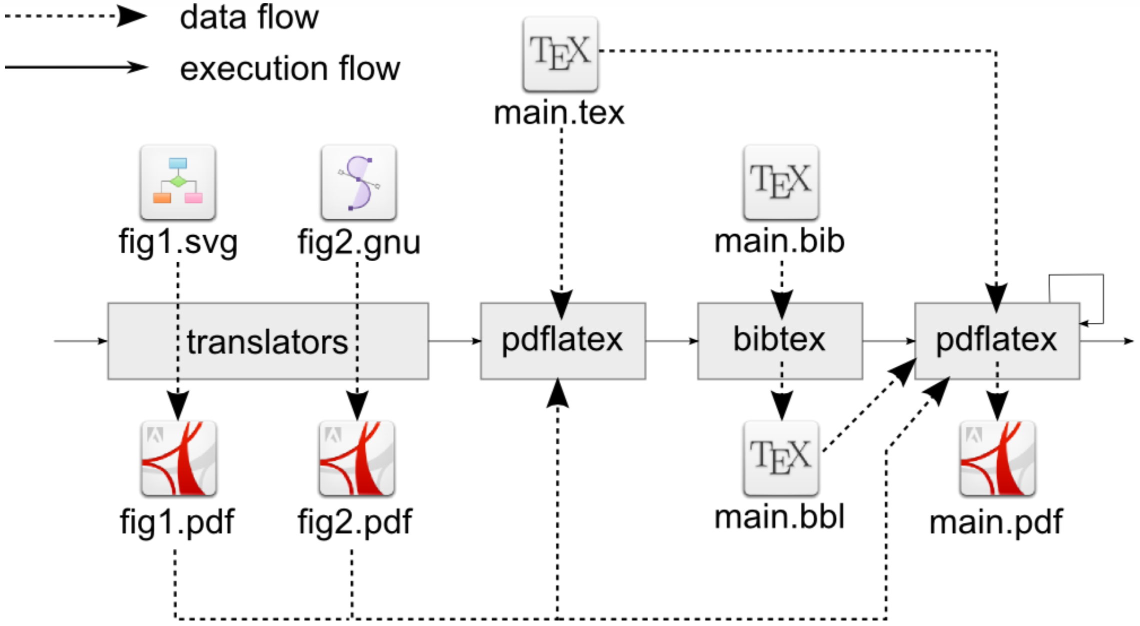
→ LaTeX is a compiled language: you must use a **compiler** to create the final ".pdf" from the ".tex" source file

→ Originally, a latex source ".tex" was transformed to DVI then to PostScript or PDF

→ The **pdflatex** compiler creates the final PDF from the source file directly

→ To use Unicode text and advanced functions, **xelatex** and **lualatex** are modern compilers

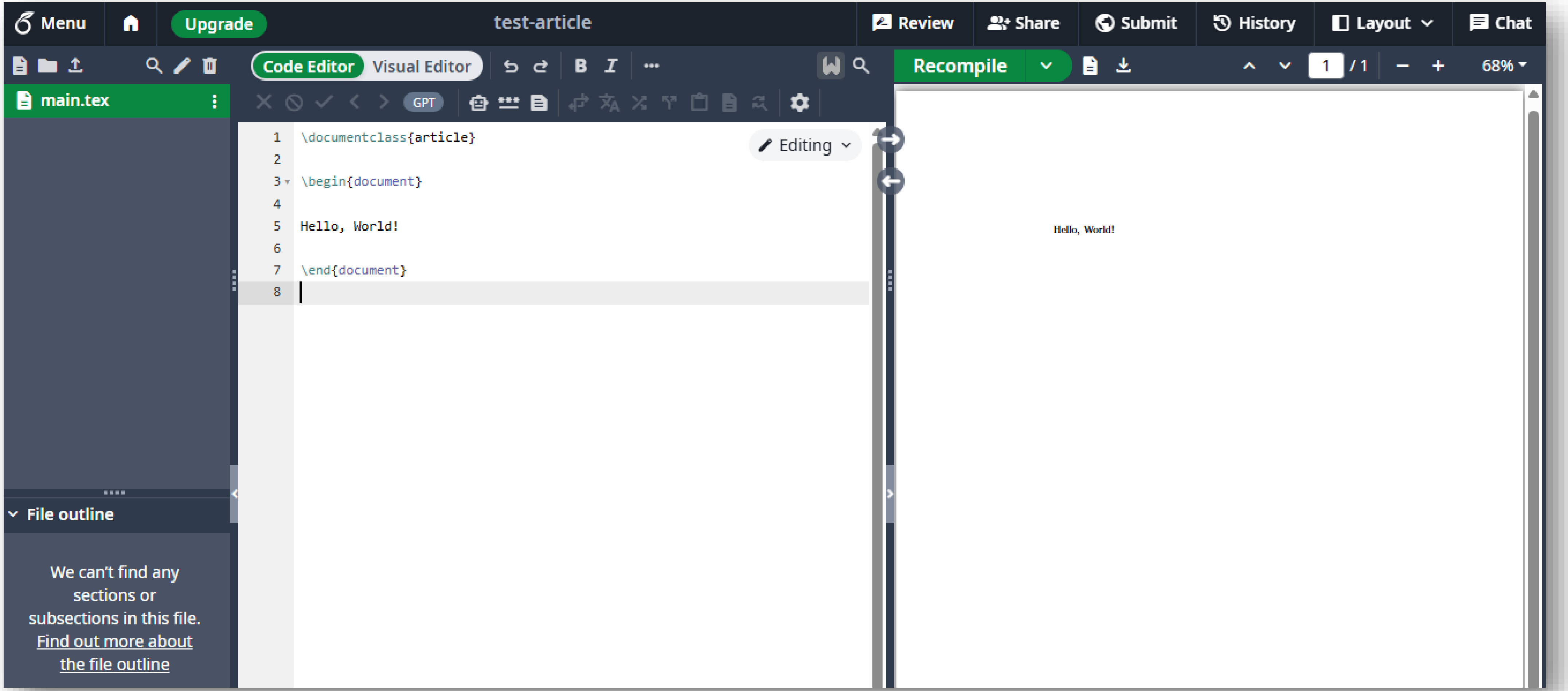




Source: https://www.overleaf.com/learn/latex/Choosing_a_LaTeX_Compiler%23Other_compilers, https://www.arakhne.org/autolatex/

4. LaTeX first article

Using overleaf



Source: https://www.overleaf.com/

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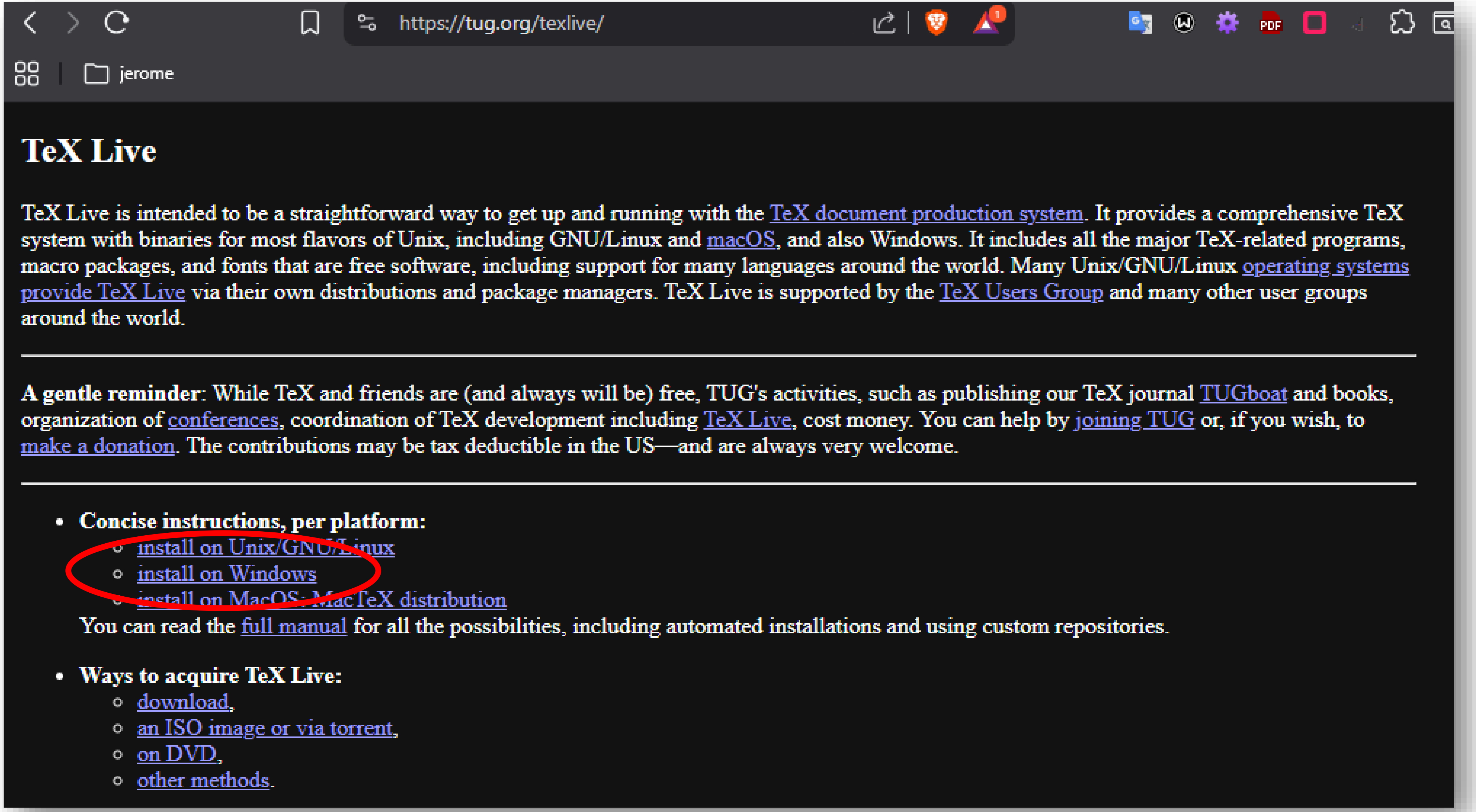


TeX Live

TeX Live is intended to be a straightforward way to get up and running with the TeX document production system. It provides a comprehensive TeX system with binaries for most flavors of Unix, including GNU/Linux and macOS, and also Windows. It includes all the major TeX-related programs, macro packages, and fonts that are free software, including support for many languages around the world. Many Unix/GNU/Linux operating systems provide TeX Live via their own distributions and package managers. TeX Live is supported by the TeX Users Group and many other user groups around the world.

A gentle reminder: While TeX and friends are (and always will be) free, TUG's activities, such as publishing our TeX journal TUGboat and books, organization of conferences, coordination of TeX development including TeX Live, cost money. You can help by joining TUG or, if you wish, to make a donation. The contributions may be tax deductible in the US—and are always very welcome.

- **Concise instructions, per platform:**
  - install on Unix/GNU/Linux
  - install on Windows
  - install on MacOS: MacTeX distribution

  You can read the full manual for all the possibilities, including automated installations and using custom repositories.
- **Ways to acquire TeX Live:**
  - download,
  - an ISO image or via torrent,
  - on DVD,
  - other methods.

Using VSCode and TexLive on Windows

4. LaTeX first article

TeX Live on Windows

Contents

- Easy install
- Other options
- 64- and 32-bit binaries
- Perl and Tcl/Tk on Windows
- Multiple TeX distributions
- Security
- Administrator privileges
- Network installations and the TeX Live launcher
- Fixing permission problems, especially with texmf-local
- Non-ascii characters in TEXDIR
- Troubleshooting other problems

Easy install

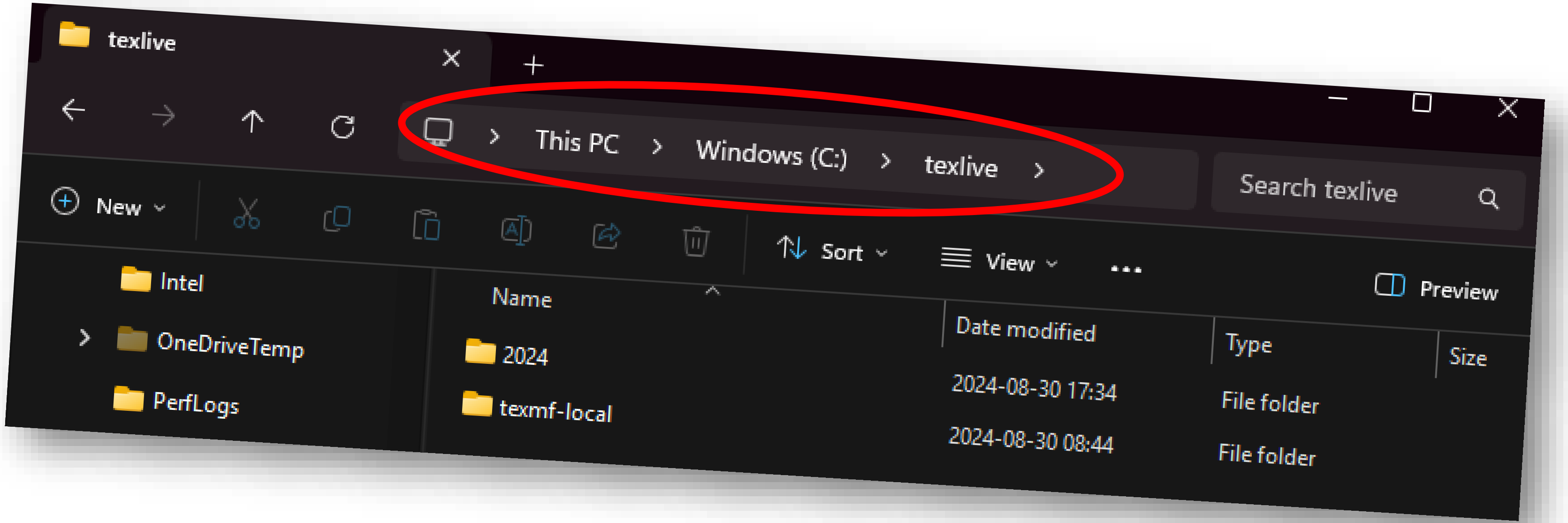
When installing from the internet, we recommend downloading and running install-tl-windows.exe.

This installer first unpacks itself and then starts the installer proper, which is the same as for other platforms. An 'Advanced' button gives you many

Using VSCode and TexLive on Windows

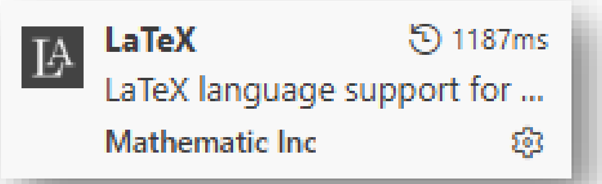
4. LaTeX first article

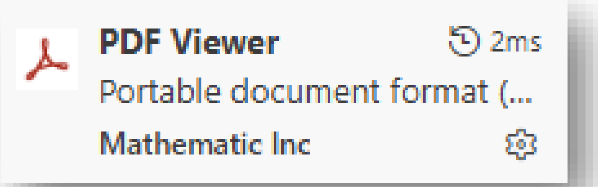
Using VSCode and TexLive on Windows

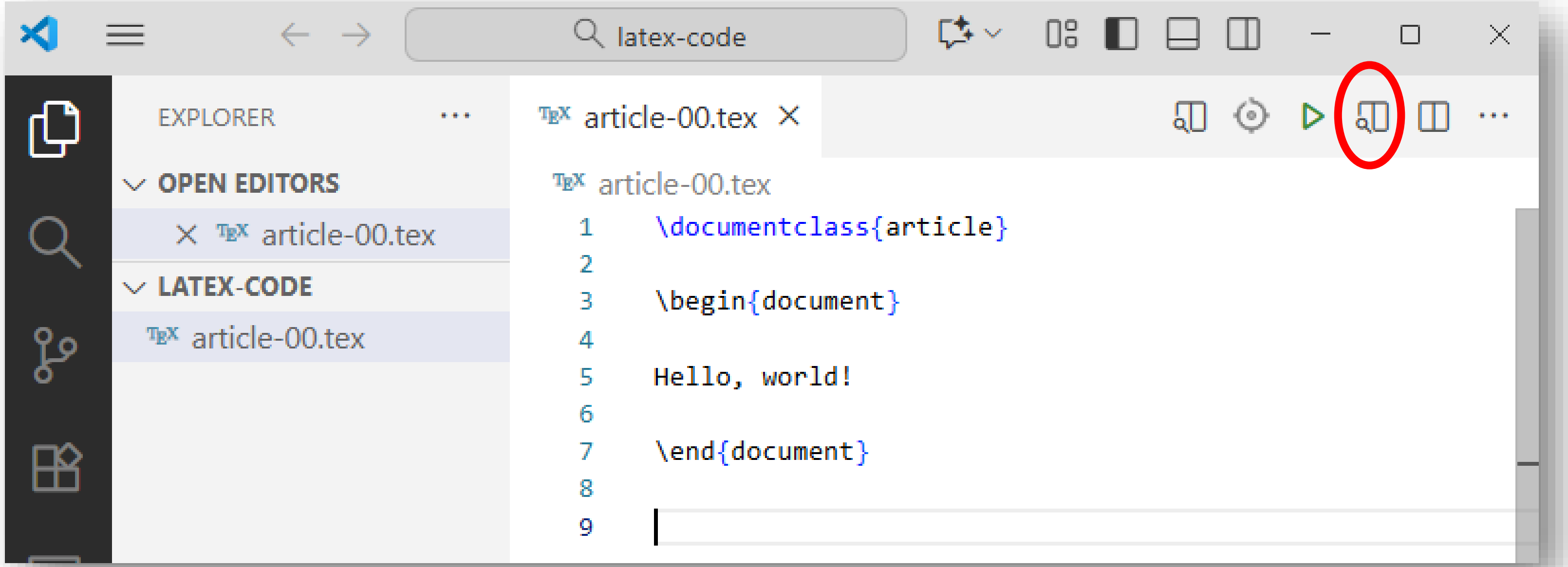


4. LaTeX first article

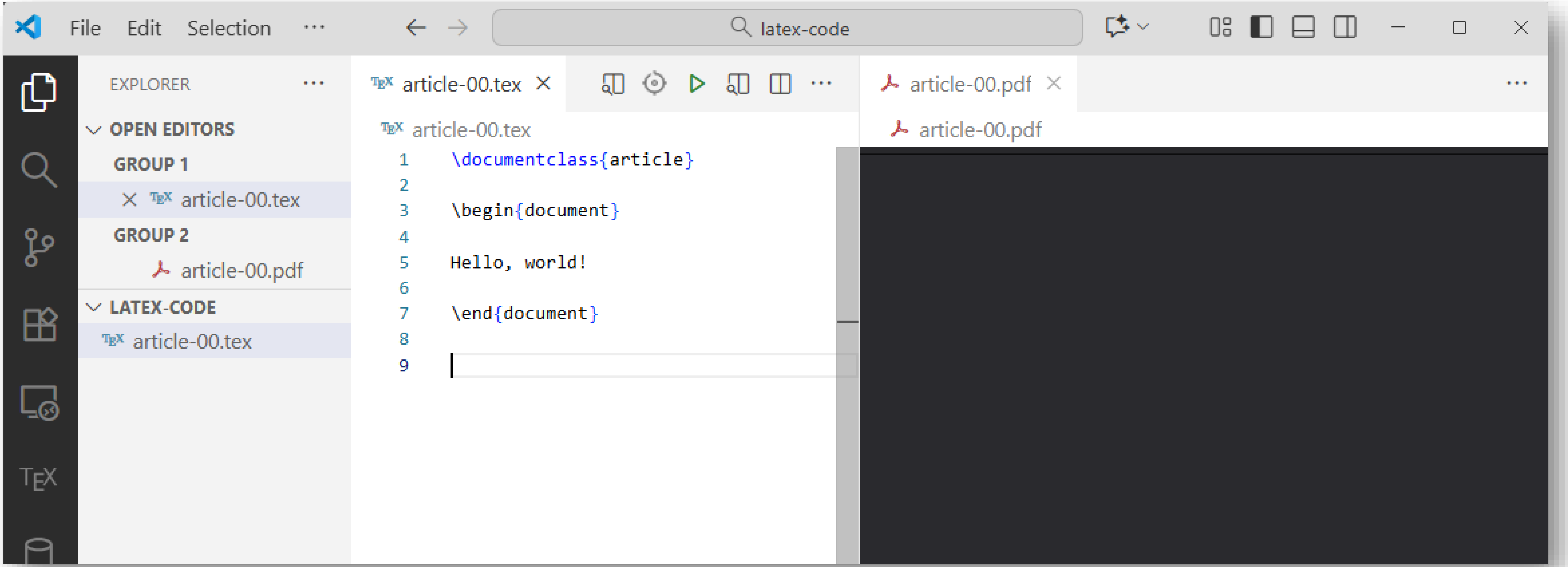
Using VSCode and TexLive on Windows







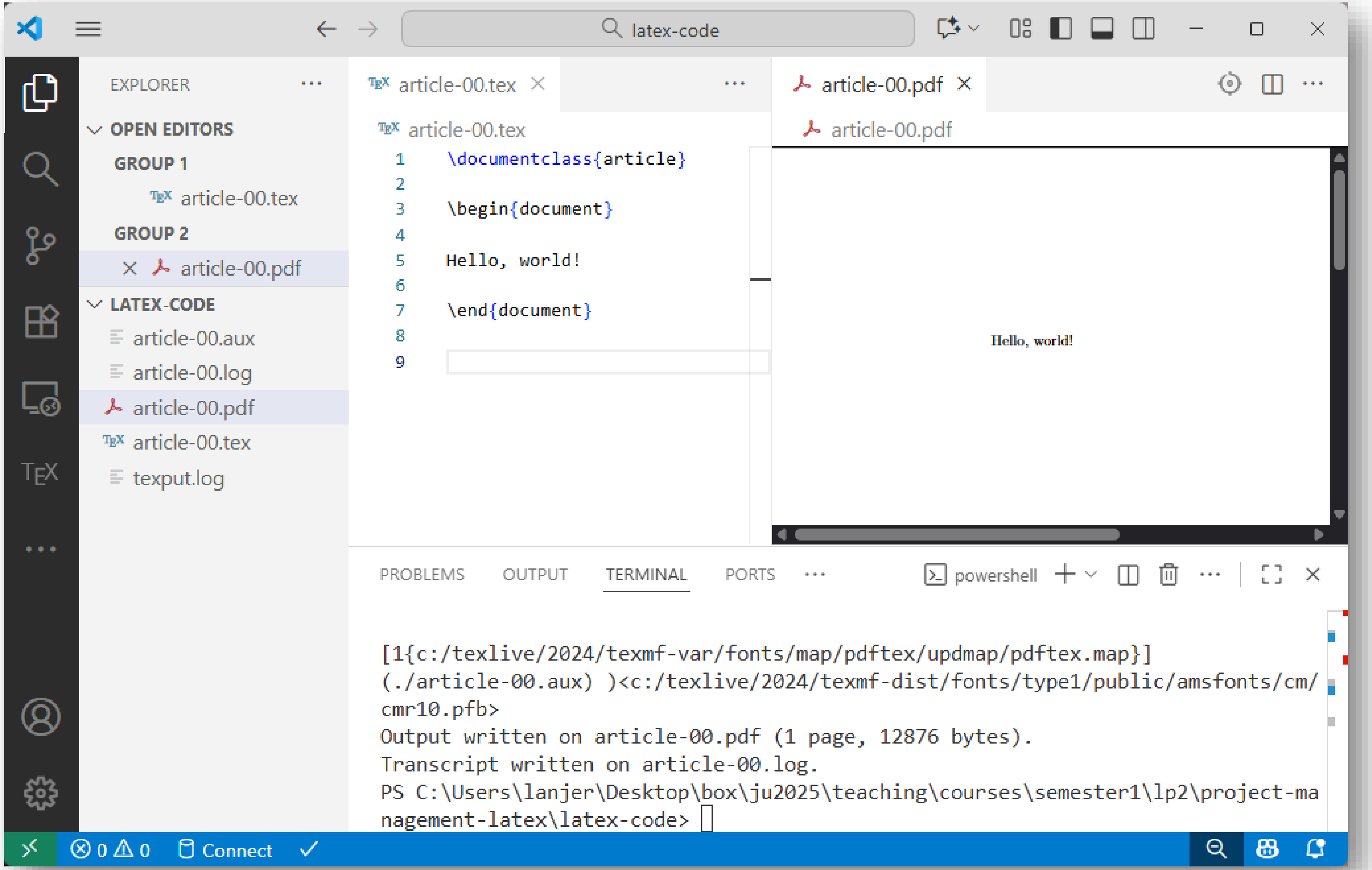
4. LaTeX first article



4. LaTeX first article

```
PS C:\Users\lanjer\Desktop\box\ju2025\teaching\courses\semester1\lp2\project-management-latex\latex-code> pdflatex.exe article-00.tex
This is pdfTeX, Version 3.141592653-2.6-1.40.26 (TeX Live 2024) (preloaded format=pdflatex)
 restricted \write18 enabled.
entering extended mode
(./article-00.tex
LaTeX2e <2024-06-01> patch level 2
L3 programming layer <2024-08-16>
(c:/texlive/2024/texmf-dist/tex/latex/base/article.cls
Document Class: article 2024/02/08 v1.4n Standard LaTeX document class
(c:/texlive/2024/texmf-dist/tex/latex/base/size10.clo))
(c:/texlive/2024/texmf-dist/tex/latex/l3backend/l3backend-pdftex.def)
No file article-00.aux.

[1{c:/texlive/2024/texmf-var/fonts/map/pdftex/updmap/pdftex.map}]
(./article-00.aux) )<c:/texlive/2024/texmf-dist/fonts/type1/public/amsfonts/cm/
cmr10.pfb>
Output written on article-00.pdf (1 page, 12876 bytes).
Transcript written on article-00.log.
PS C:\Users\lanjer\Desktop\box\ju2025\teaching\courses\semester1\lp2\project-management-latex\latex-code>
```

4. LaTeX first article



4. LaTeX first article

The preamble starts by defining the documentclass, it also contains the packages used in the document as well as the commands and functions used in the body.

The body contains the text, pictures and images that will appear in your final document.

preamble

```
\documentclass{article}
```

body (document)

```
\begin{document}

Hello, world!

\end{document}
```

4. LaTeX first article

The ”\documentclass” command
with one parameter ”article”

preamble

```
\documentclass{article}
```

body
(document)

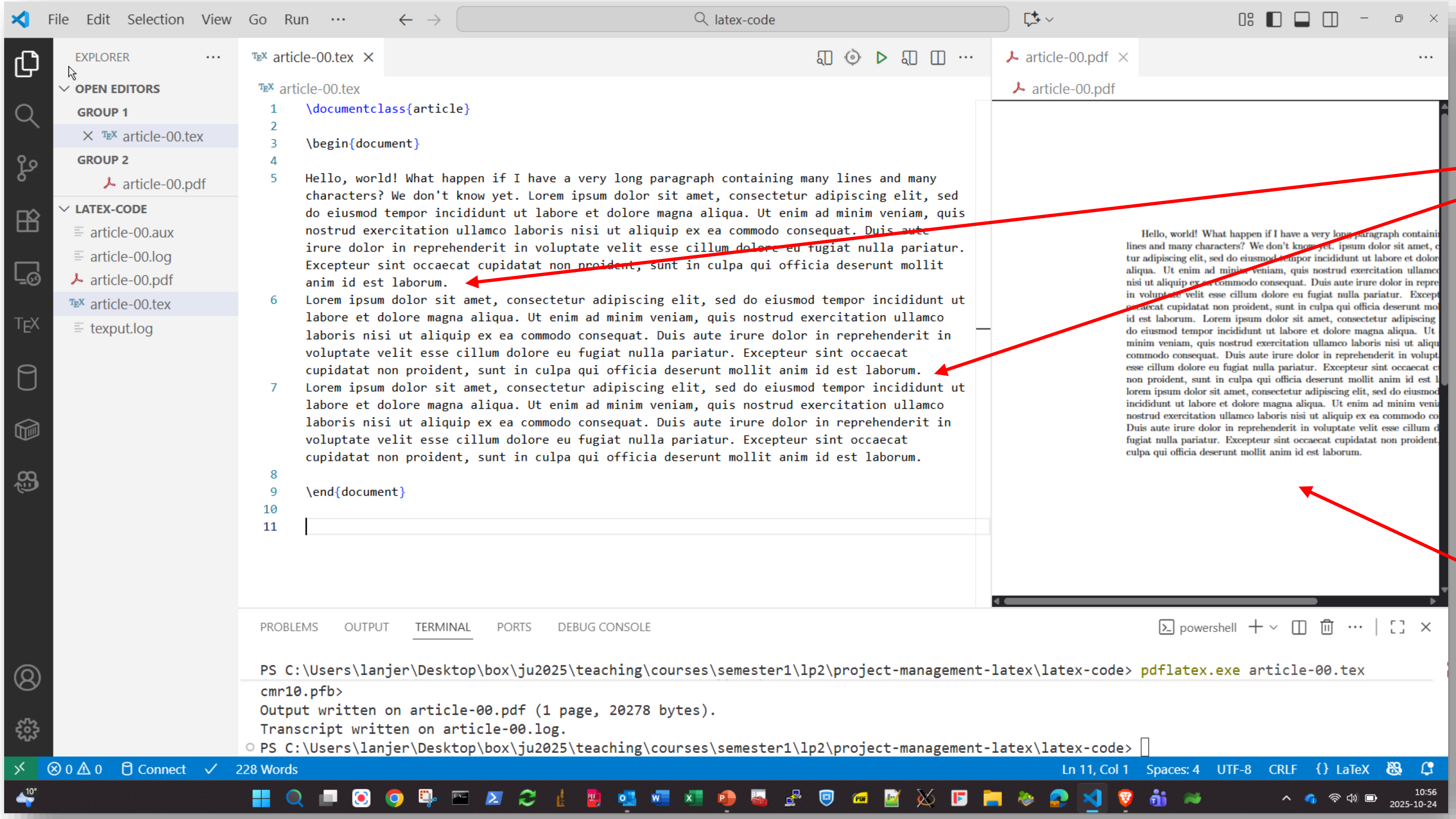
```
\begin{document}

Hello, world!

\end{document}
```

The beginning of the environment
”document”, the command ”\begin”
with one parameter ”document”

The end of the environment
”document”, the command ”\end”
with one parameter ”document”

4. LaTeX first article

What happens with a longer text?



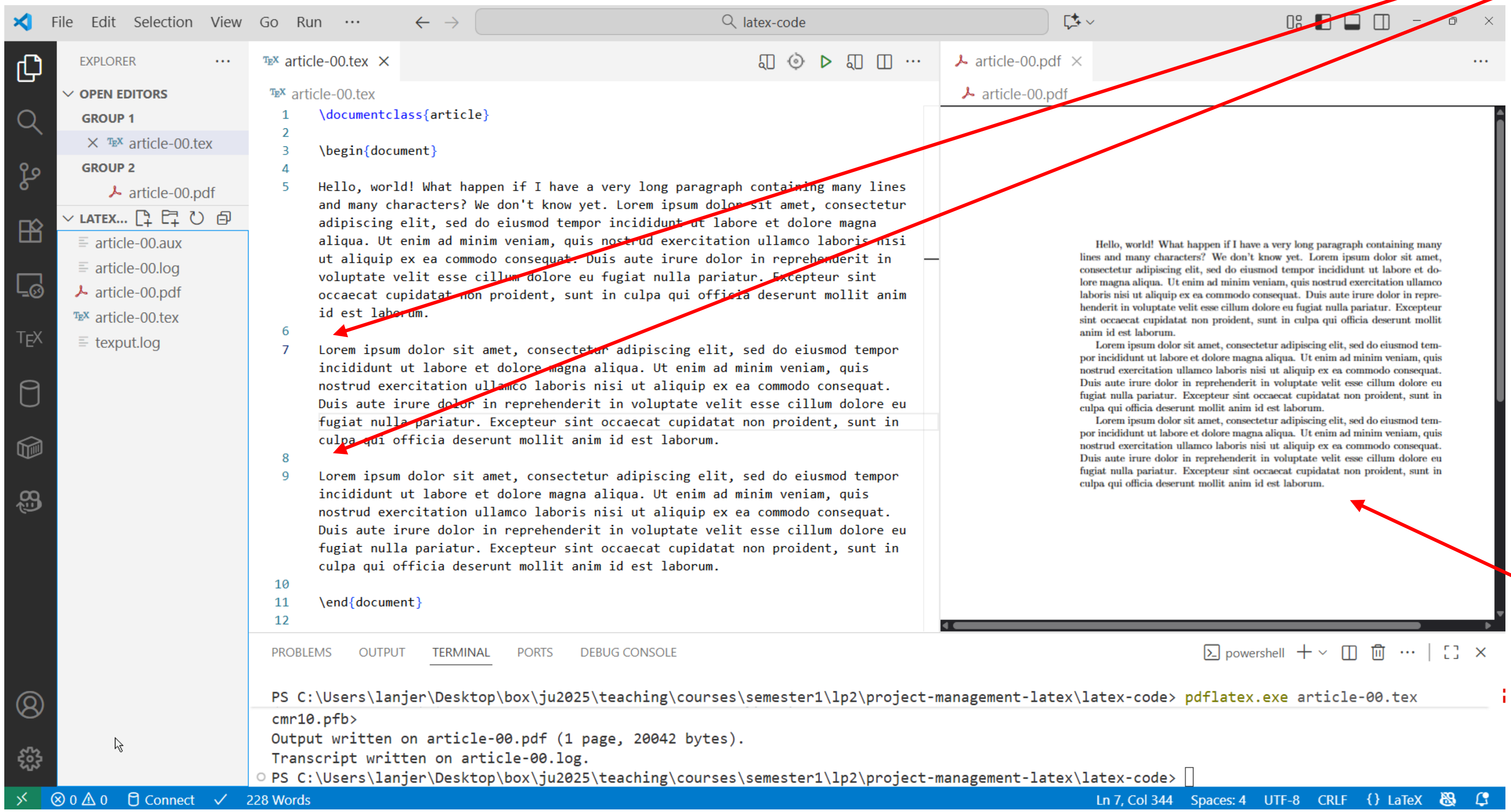
No line break, just a new line

Only one paragraph at the end!

4. LaTeX first article

What happens with a longer text?

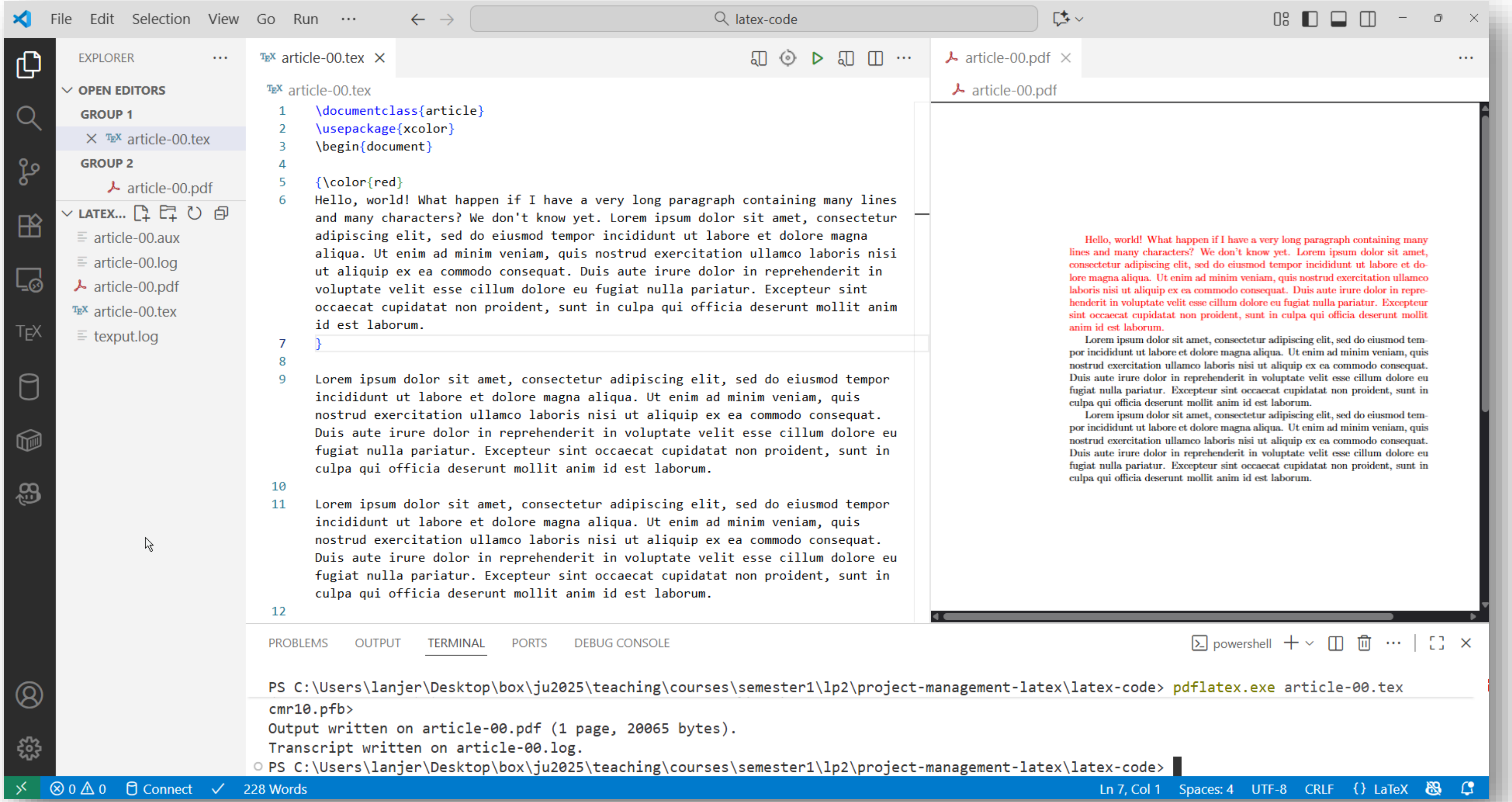
Line breaks



Three paragraphs at the end!

5. Several LaTeX commands

Adding colors to the document



5. Several LaTeX commands

Adding colors to the document

```
\documentclass{article}
\usepackage{xcolor}

\begin{document}

{\color{red}
Hello, world! What happen if I have a very long paragraph containing many lines
and many characters? We don't know yet. Lorem ipsum dolor sit amet, consectetur
adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna
aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi
ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in
voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint
occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim
id est laborum.
}

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incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis
nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.
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fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in
culpa qui officia deserunt mollit anim id est laborum.

\end{document}
```
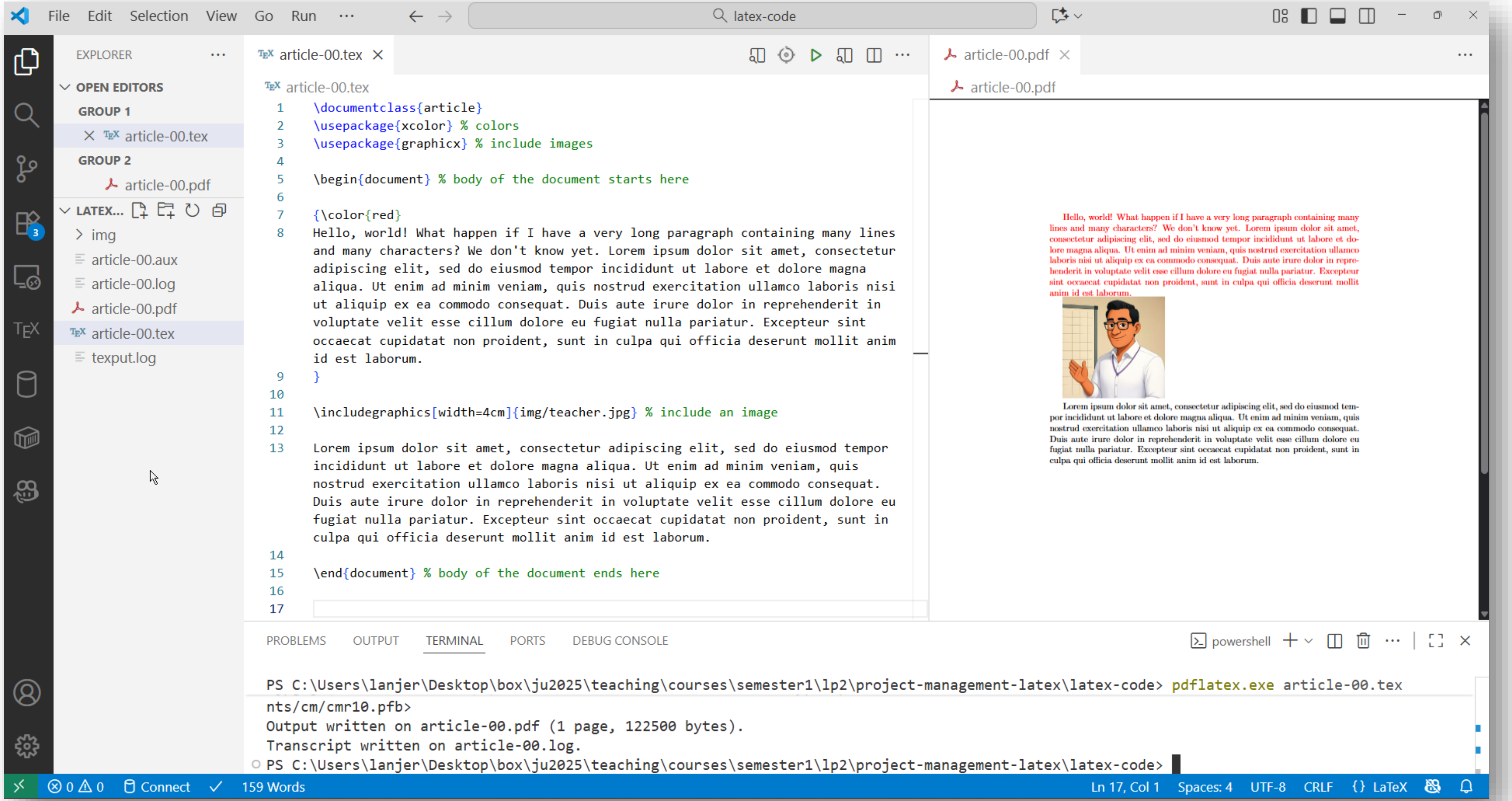

Use a pre-defined package

Apply a color to a block { }

5. Several LaTeX commands



5. Several LaTeX commands

```
\documentclass{article}
\usepackage{xcolor} % colors
\usepackage{graphicx} % include images

\begin{document} % body of the document starts here

{\color{red}
Hello, world! What happen if I have a very long paragr
and many characters? We don't know yet. Lorem ipsum do
adipiscing elit, sed do eiusmod tempor incididunt ut l
aliqua. Ut enim ad minim veniam, quis nostrud exercita
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voluptate velit esse cillum dolore eu fugiat nulla pari
occaecat cupidatat non proident, sunt in culpa qui offi
id est laborum.
}

\includegraphics[width=4cm]{img/teacher.jpg} % include an image

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor
incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis
nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.
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fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in
culpa qui officia deserunt mollit anim id est laborum.

\end{document} % body of the document ends here
```

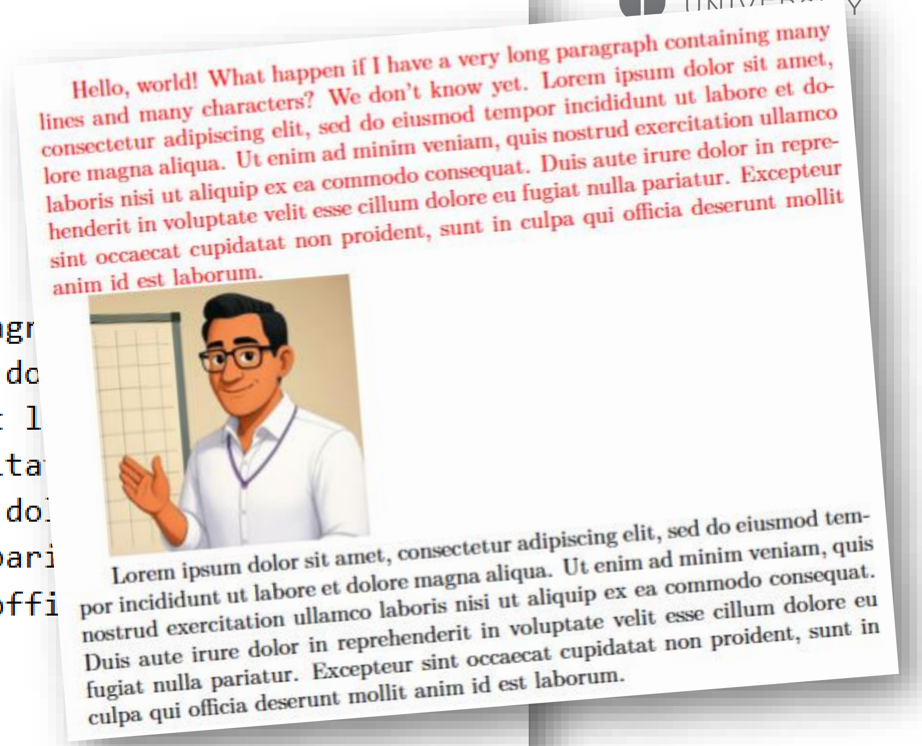

Hello, world! What happen if I have a very long paragraph containing many lines and many characters? We don't know yet. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

5. Several LaTeX commands

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```
\documentclass{article}
\usepackage{xcolor} % colors
\usepackage{graphicx} % include images

\title{My first \LaTeX document} % title
\author{Jérôme Landré} % author
\date{\today} % date

\begin{document} % body of the document starts here

\maketitle % generate title
\tableofcontents % generate table of contents

\section{Introduction} % section
\subsection{Foreword} % subsection
{\color{red}
Hello, world! What happen if I have a very long paragraph containing many lines and many
characters? We don't know yet. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed
do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam,
quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis
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pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia
deserunt mollit anim id est laborum.
}

\subsection{Why use \LaTeX?} % subsection
\LaTeX\ is a high-quality typesetting system; it includes features designed for the
production of technical and scientific documentation.

\section{My image}
\includegraphics[width=4cm]{img/teacher.jpg} % include an image

\section{Conclusion}
Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt
ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in
reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur
sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est
laborum.

\end{document} % body of the document ends here
```

My first LaTeXdocument

Jérôme Landré

October 24, 2025

Contents

1 Introduction

1.1 Foreword

Hello, world! What happen if I have a very long paragraph containing many lines and many characters? We don't know yet. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

1.2 Why use LaTeX?

LaTeX is a high-quality typesetting system; it includes features designed for the production of technical and scientific documentation.

2 My image

3 Conclusion

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

5. Several LaTeX commands

Adding Mathematics!

```
\section{Introduction} % section
\subsection{Foreword} % subsection

Let's study a simple {\color{red} derivative} in \LaTeX:

$$ \frac{d}{dx} (3 x^2 - \sin x )= 6 x - \cos x $$

And a formula to compute the value of $\pi$:

$$ \pi = 4 \int_{0}^{1} \frac{1}{1+x^2} \; dx. $$
```

1 Introduction

1.1 Foreword

Let's study a simple derivative in LaTeX:

$$\frac{d}{dx}(3x^2 - \sin x) = 6x - \cos x$$

And a formula to compute the value of π:

$$\pi = 4 \int_0^1 \frac{1}{1+x^2} \, dx.$$

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```
\newpage
\subsection{Why use \LaTeX{}?} % subsection
\LaTeX\ is a high-quality typesetting system; it includes features designed for the production of
technical and scientific documentation.

\section{My image}

This is an image of a teacher:

\includegraphics[width=4cm]{img/teacher.jpg} % include an image

You can see in this image a \textbf{teacher} \underline{in front of} a \textit{whiteboard}.
```

An optional parameter:
[name=value]

1.2 Why use LaTeX?

LaTeX is a high-quality typesetting system; it includes features designed for the production of technical and scientific documentation.

2 My image

This is an image of a teacher:

You can see in this image a **teacher** in front of a *whiteboard*.

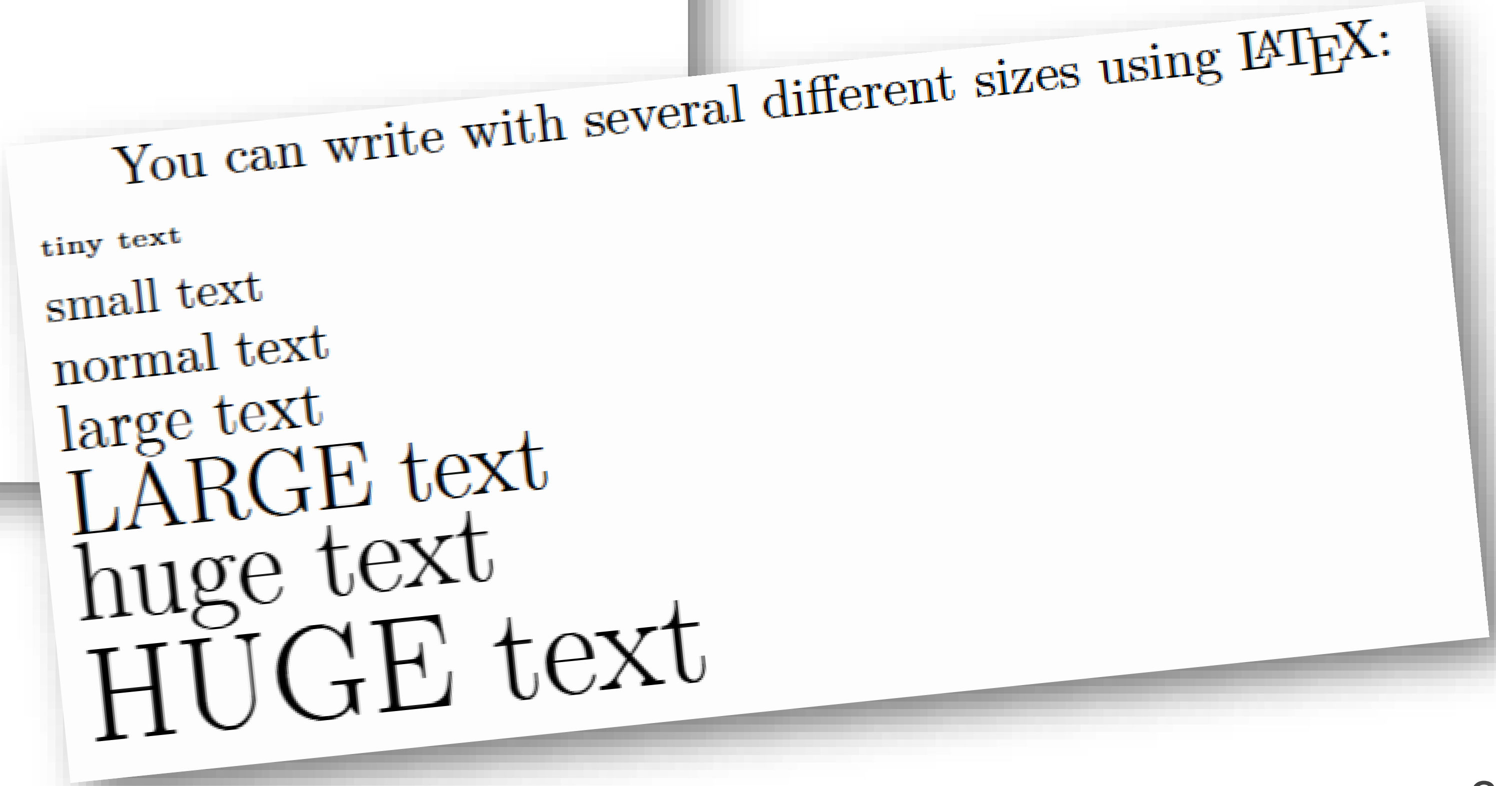
Opening a block

Setting the paragraph indentation to 0
(only inside **THIS** block)

```
You can write with several different sizes using \LaTeX{}:

{\parindent=0pt
{\tiny tiny text} \\
{\small small text} \\
{\normalsize normal text} \\
{\large large text} \\
{\LARGE LARGE text} \\
{\huge huge text} \\
{\Huge HUGE text} \\
}
```

Closing the block



```latex
\section{Using tables}

In \LaTeX{}, you can use the tabular environment to create tables:

\medskip
\begin{tabular}{|l|c|r|}
\hline
Teacher & Course & Number of students \\
\hline
Bilal Z. & Research Methods & 90 \\
\hline
Jérôme L. & Databases & 80 \\
\hline
Jérôme L. & Operating Systems & 50 \\
\hline
\end{tabular}
```

3 Using tables

In LaTeX, you can use the tabular environment to create tables:

Teacher	Course	Number of students
Bilal Z.	Research Methods	90
Jérôme L.	Databases	80
Jérôme L.	Operating Systems	50

```
\newpage
\section{Using figures\dots}

You can also declare a figure, use a caption and a label to reference it later (but you need to compile the file TWICE to get the correct reference).

{\parindent=0pt
\begin{figure}[htbp]
\centering
\includegraphics[width=12cm]{img/arbre_lac_uchon1.jpg}
\caption{A lake in Uchon, Burgundy, France.}
\label{fig:lake}
\end{figure}
}
```

4 Using figures...

You can also declare a figure, use a caption and a label to reference it later (but you need to compile the file TWICE to get the correct reference).

Figure 1: A lake in Uchon, Burgundy, France.

```latex
\section{Using lists}

\subsection{The itemize environment}

You can create itemized lists:
\begin{itemize}
\item Bilal
\item Mira
\item Jasmin
\item Jérôme
\item Maria
\item \dots
\end{itemize}
```

5 Using lists

5.1 The itemize environment

You can create itemized lists:

- Bilal
- Mira
- Jasmin
- Jérôme
- Maria
- ...

```latex
\subsection{The enumerate environment}
You can create enumerated lists:
\begin{enumerate}
\item The winner
\item The second
\item The third
\item The fourth
\item \dots
\end{enumerate}
```

5.2 The enumerate environment

You can create enumerated lists:

1. The winner
2. The second
3. The third
4. The fourth
5. ...

```latex
\newpage
\section{Conclusion}

In figure \ref{fig:lake}, you can see a lake in Uchon, Burgundy, France. It is my preferred place to
relax when I am home. It is so quiet and peaceful! If you want, you can add the page of the figure like
this: see figure \ref{fig:lake} on page \pageref{fig:lake}.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et
dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip
ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu
fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia
deserunt mollit anim id est laborum.

\end{document} % body of the document ends here
```

6 Conclusion

In figure 1, you can see a lake in Uchon, Burgundy, France. It is my preferred place to relax when I am home. It is so quiet and peaceful! If you want, you can add the page of the figure like this: see figure 1 on page 3.

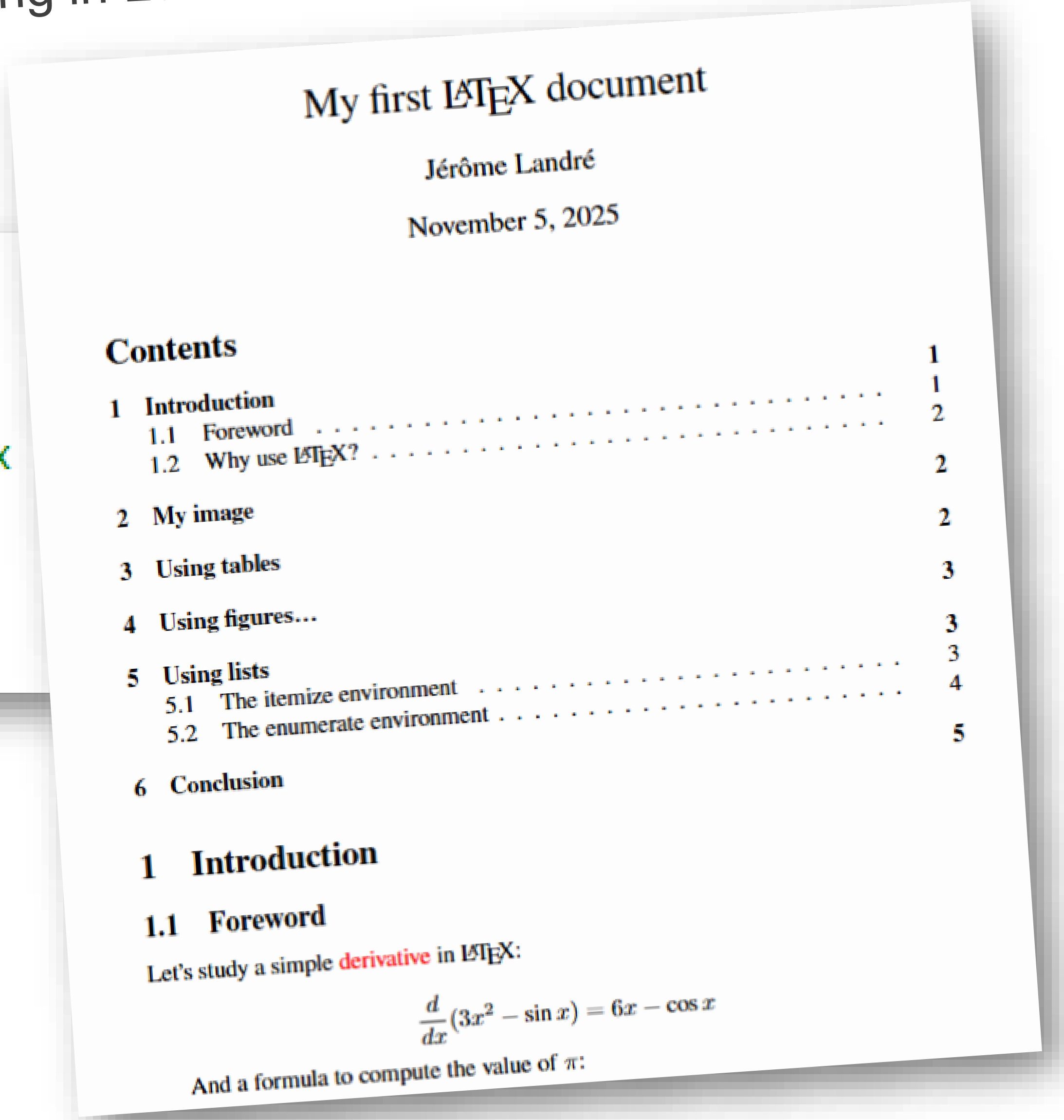
Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

6. LuaLaTeX

Dealing with Unicode characters, fonts,
Programming in Lua and much more…

```
\documentclass{article}
\usepackage{xcolor} % colors
\usepackage{graphicx} % include images

\usepackage{fontspec} % font selection with LuaLaTeX
%\setmainfont{Cantarell} % sans-serif font
%\setmainfont{Cabin} % sans-serif font
%\setmainfont{Georgia} % serif font
\setmainfont{FreeSerif} % serif font
```



My first LaTeX document

Jérôme Landré

November 5, 2025

Contents

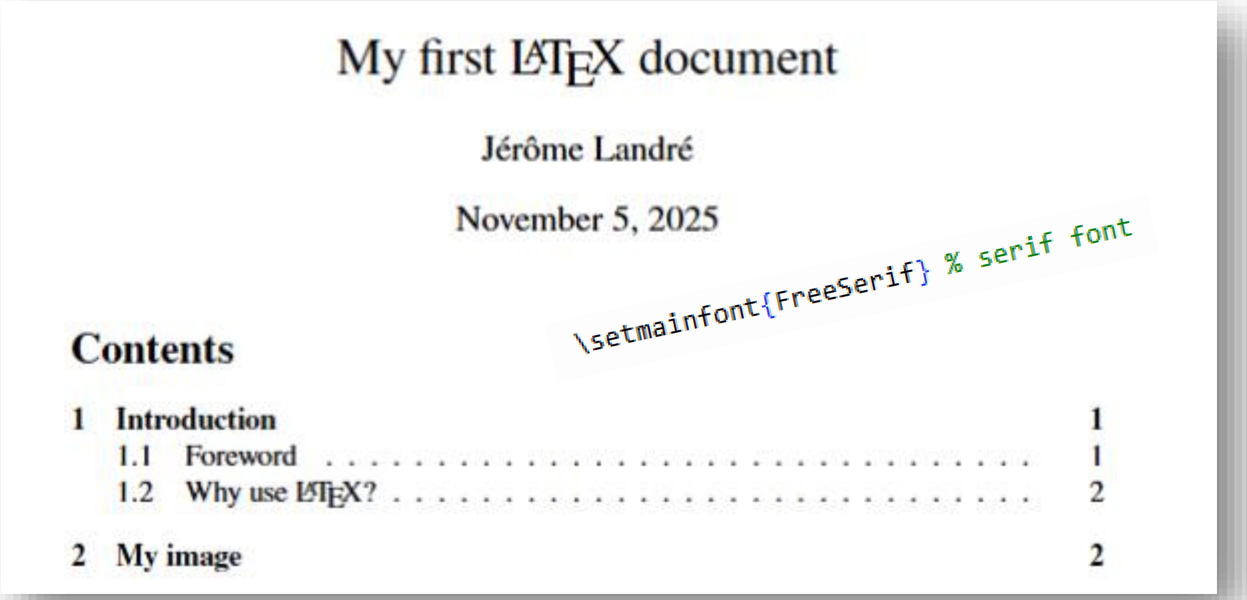
1 Introduction

1.1 Foreword

Let's study a simple derivative in LaTeX:

$$\frac{d}{dx}(3x^2 - \sin x) = 6x - \cos x$$

And a formula to compute the value of π:


My first LaTeX document

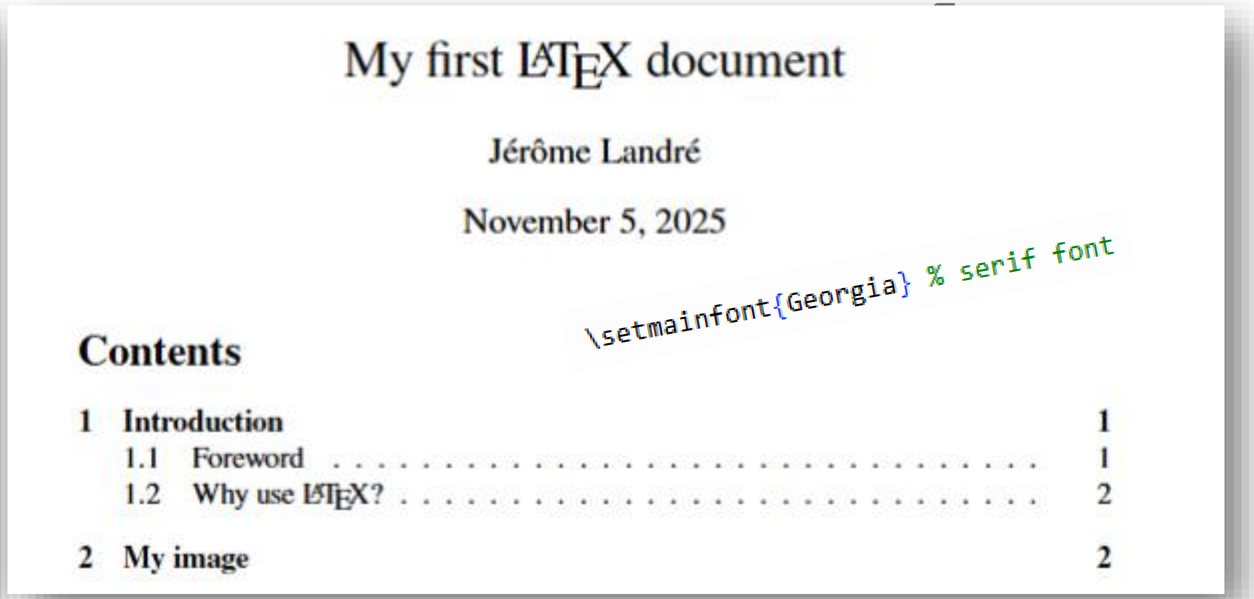
Jérôme Landré

November 5, 2025

```
\setmainfont{FreeSerif} % serif font
```

Contents

1	**Introduction**	1
1.1	Foreword	1
1.2	Why use LaTeX?	2
2	**My image**	2


My first LaTeX document

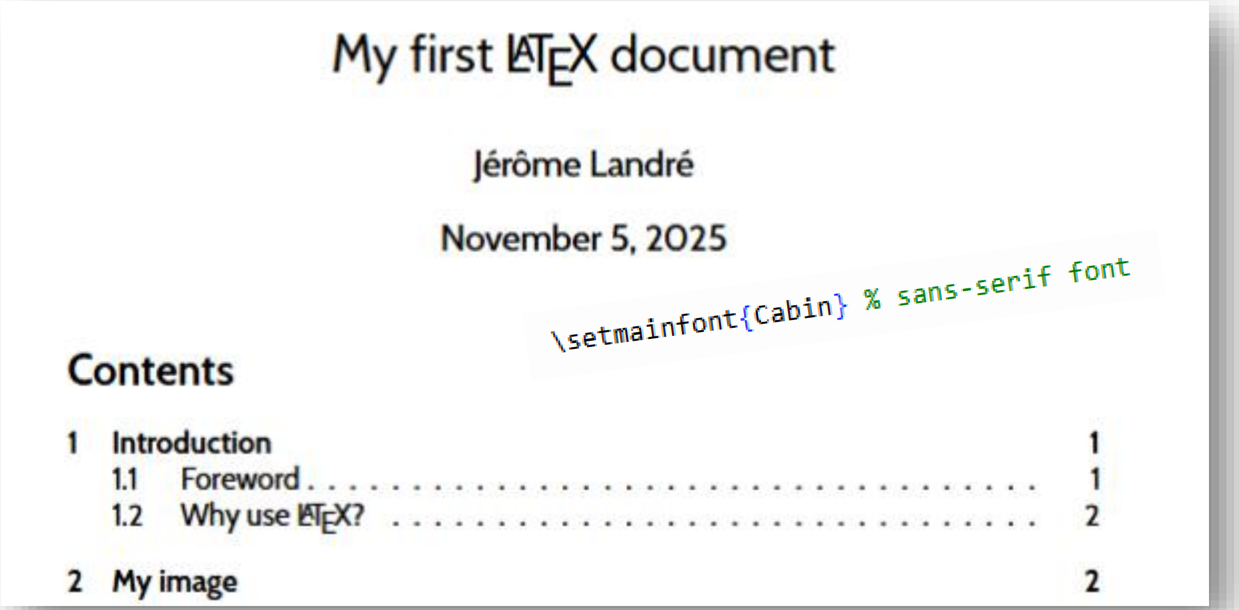
Jérôme Landré

November 5, 2025

```
\setmainfont{Georgia} % serif font
```

Contents

1	**Introduction**	1
1.1	Foreword	1
1.2	Why use LaTeX?	2
2	**My image**	2


My first LaTeX document

Jérôme Landré

November 5, 2025

```
\setmainfont{Cabin} % sans-serif font
```

Contents

1	**Introduction**	1
1.1	Foreword	1
1.2	Why use LaTeX?	2
2	**My image**	2


My first LaTeX document

Jérôme Landré

November 5, 2025

```
\setmainfont{Cantarell} % sans-serif font
```

Contents

1	**Introduction**	1
1.1	Foreword	1
1.2	Why use LaTeX?	2
2	**My image**	2

JÖNKÖPING UNIVERSITY

6. LuaLaTeX

Dealing with Unicode characters, fonts,
Programming in Lua and much more…

```
fonts
    Literata.ttf
    NotoSans.ttf
```

```latex
\documentclass{article}
\usepackage{xcolor} % colors
\usepackage{graphicx} % include images

\usepackage{fontspec} % font selection with LuaLaTeX

\setmainfont{Literata}[
    Path = ./fonts/,
    Extension = .ttf
]

\title{My first \LaTeX document} % title
\author{Jérôme Landré} % author
\date{\today} % date

\begin{document} % body of the document starts here

\maketitle % generate title
\tableofcontents % generate table of contents
```

1 Introduction

1.1 Foreword

Let's study a simple derivative in LaTeX:

$$\frac{d}{dx}(3x^2 - \sin x) = 6x - \cos x$$

And a formula to compute the value of π:

$$\pi = 4\int_0^1 \frac{1}{1+x^2}\,dx.$$

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6. LuaLaTeX

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Programming in Lua and much more…

- fonts
  - Literata.ttf
  - NotoSans.ttf

```
\documentclass{article}
\usepackage{xcolor} % colors
\usepackage{graphicx} % include images

\usepackage{fontspec} % font selection with LuaLa

\setmainfont{NotoSans}[
    Path = ./fonts/,
    Extension = .ttf
]

\title{My first \LaTeX document} % title
\author{Jérôme Landré} % author
\date{\today} % date

\begin{document} % body of the document starts he

\maketitle % generate title
\tableofcontents % generate table of contents
```

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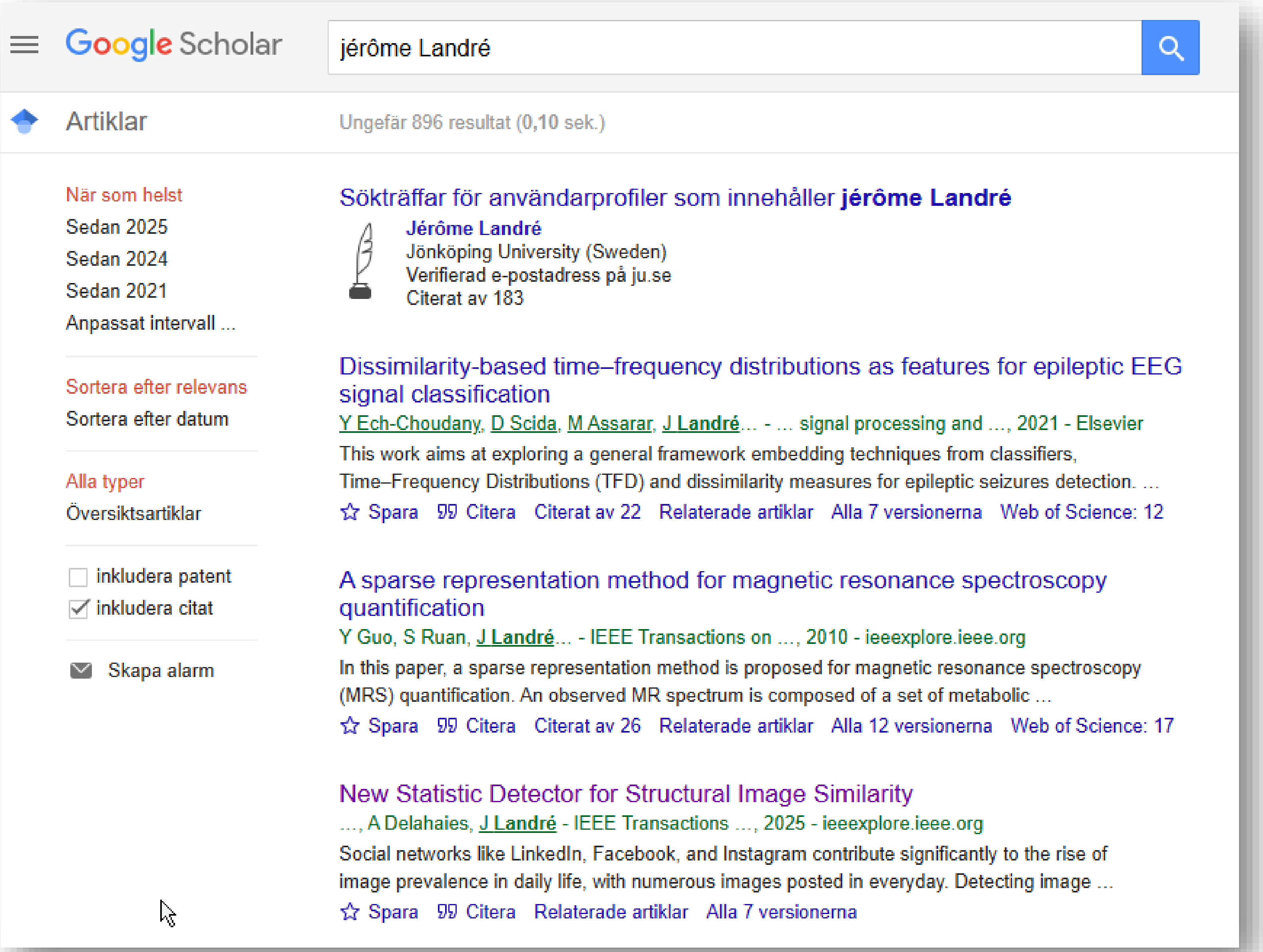
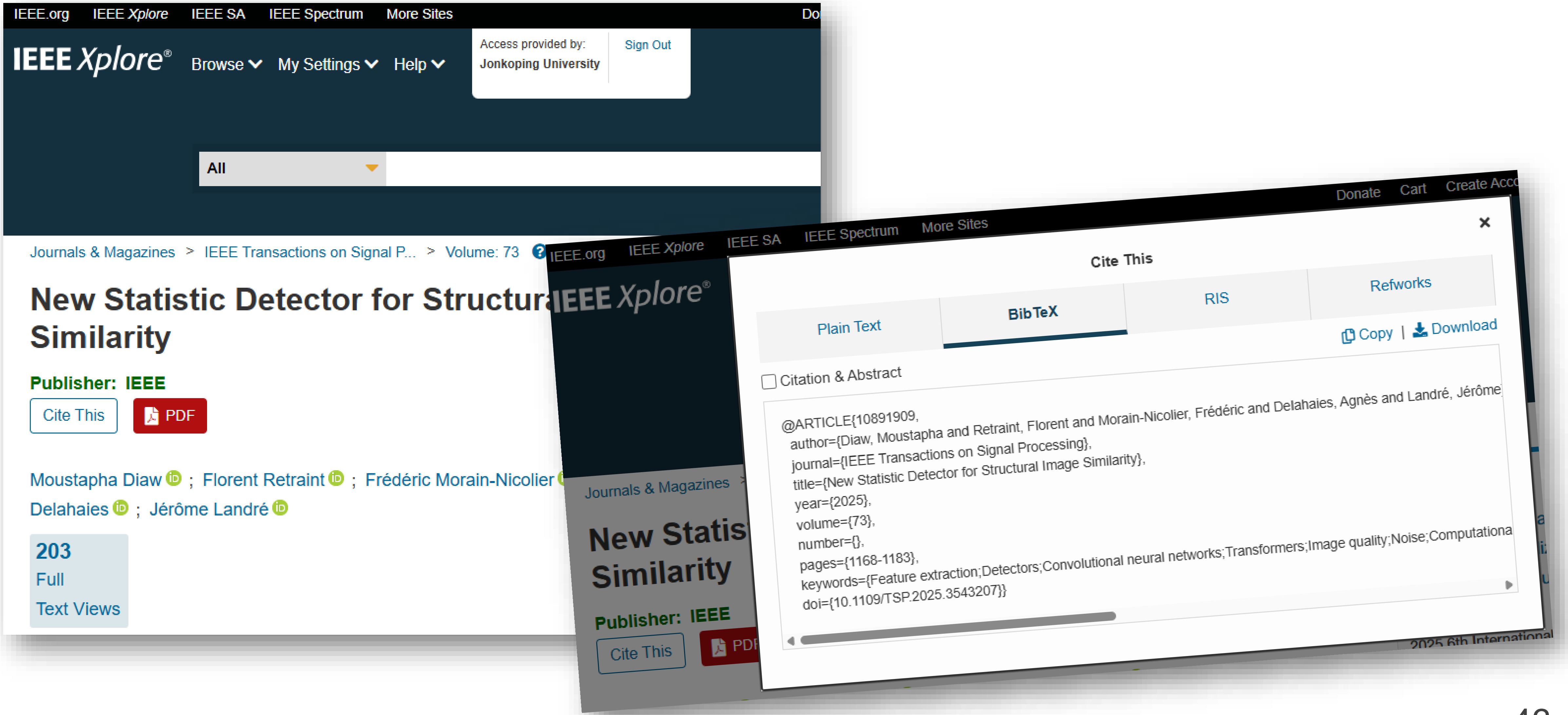
$$\pi = 4\int_0^1 \frac{1}{1+x^2}\,dx.$$

7. Bibliography

- LaTeX can manage your references (the bibliography) for you!



JÖNKÖPING UNIVERSITY



7. Bibliography

mybiblio.bib

mybiblio.bib > ...

```
@ARTICLE{10891909,
  author={Diaw, Moustapha and Retraint, Florent and Morain-Nicolier, Frédéric and Delahaies, Agnès and Landré, Jérôme},
  journal={IEEE Transactions on Signal Processing},
  title={New Statistic Detector for Structural Image Similarity},
  year={2025},
  volume={73},
  number={},
  pages={1168-1183},
  keywords={Feature extraction;Detectors;Convolutional neural networks;Transformers;Image quality;Noise;Computational modeling;Indexes;Image classification;Training;Classification;local dissimilarity map;weibull distribution;GLRT;similarity measures;pre-trained CNN},
  doi={10.1109/TSP.2025.3543207}}
```

article-05.tex > ...

```
\documentclass{article}
\usepackage{xcolor} % colors
\usepackage{graphicx} % include images

\usepackage{fontspec} % font selection with LuaLaTeX
\setmainfont{Cantarell} % sans-serif font
%\setmainfont{Cabin} % sans-serif font
%\setmainfont{Georgia} % serif font
%\setmainfont{FreeSerif} % serif font

\usepackage[backend=biber, style=authoryear]{biblatex}
\addbibresource{mybiblio.bib} % Your .bib file

```

```
\newpage
\subsection{Why use \LaTeX{}?} % subsection
\LaTeX\ is a high-quality typesetting system; it includes features designed
for the production of technical and scientific documentation.

In the article \cite{10891909}, the authors used \LaTeX{} to write their
research paper.
```

```
\newpage
\section{Conclusion}

In figure \ref{fig:lake}, you can see a lake in Uchon, Burgundy, France. It is my preferred
place to relax when I am home. It is so quiet and peaceful! If you want, you can add the
page of the figure like this: see figure \ref{fig:lake} on page \pageref{fig:lake}.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt
ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in
reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur
sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est
laborum.

\section{References}
\printbibliography[title=none]

\end{document} % body of the document ends here
```

7. Bibliography

PROBLEMS OUTPUT TERMINAL PORTS DEBUG CONSOLE

powershell

```
PS C:\Users\lanjer\Desktop\box\ju2025\teaching\courses\semester1\lp2\project-management-latex\latex-code> lualatex ./article-05.tex
/fonts/type1/public/amsfonts/cm/cmr7.pfb><c:/texlive/2024/texmf-dist/fonts/type
1/public/amsfonts/cm/cmsy10.pfb>
Output written on article-05.pdf (5 pages, 611003 bytes).
Transcript written on article-05.log.
PS C:\Users\lanjer\Desktop\box\ju2025\teaching\courses\semester1\lp2\project-management-latex\latex-code> lualatex ./article-05.tex;biber article-05;lualatex ./article-05.tex
```

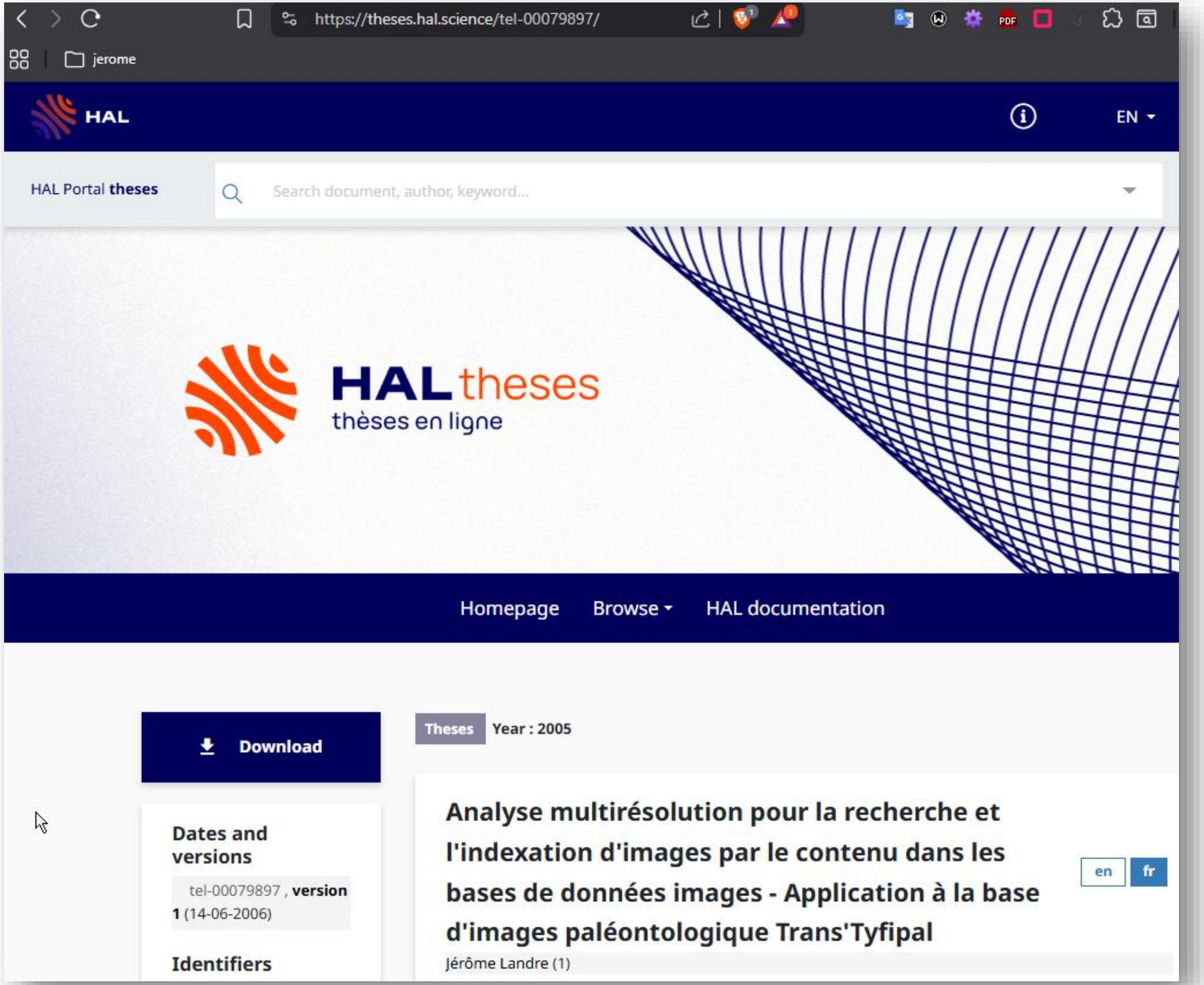
1.2 Why use LaTeX?

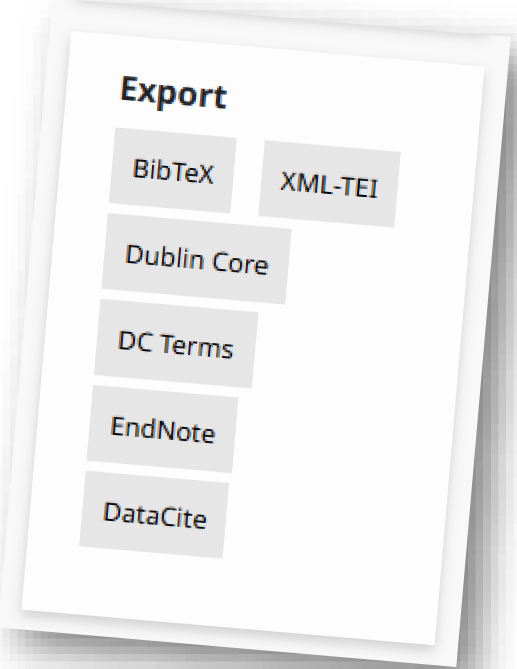
LaTeX is a high-quality typesetting system; it includes features designed for the production of technical and scientific documentation.

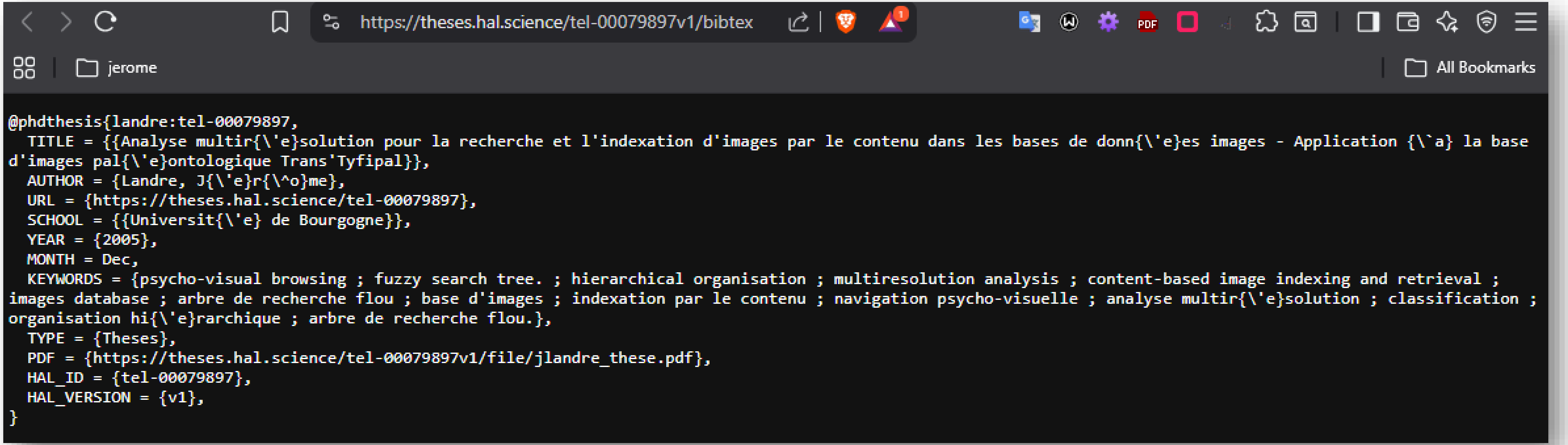
In the article Diaw et al. 2025, the authors used LaTeX to write their research paper.

7 References

Diaw, Moustapha et al. (2025). "New Statistic Detector for Structural Image Similarity". In: IEEE Transactions on Signal Processing 73, pp. 1168–1183. DOI: 10.1109/TSP.2025.3543207.


https://theses.hal.science/tel-00079897/
jerome
HAL
EN
HAL Portal theses
Search document, author, keyword...
HAL theses
thèses en ligne
Homepage
Browse
HAL documentation
Download
Theses
Year : 2005
Analyse multirésolution pour la recherche et l'indexation d'images par le contenu dans les bases de données images - Application à la base d'images paléontologique Trans'Tyfipal
en
fr
Jérôme Landre (1)
Dates and versions
tel-00079897 , version 1 (14-06-2006)
Identifiers


Export
BibTeX
XML-TEI
Dublin Core
DC Terms
EndNote
DataCite




```
@phdthesis{landre:tel-00079897,
  TITLE = {{Analyse multir{\'e}solution pour la recherche et l'indexation d'images par le contenu dans les bases de donn{\'e}es images - Application {\`a} la base d'images pal{\'e}ontologique Trans'Tyfipal}},
  AUTHOR = {Landre, J{\'e}r{\^o}me},
  URL = {https://theses.hal.science/tel-00079897},
  SCHOOL = {{Universit{\'e} de Bourgogne}},
  YEAR = {2005},
  MONTH = Dec,
  KEYWORDS = {psycho-visual browsing ; fuzzy search tree. ; hierarchical organisation ; multiresolution analysis ; content-based image indexing and retrieval ; images database ; arbre de recherche flou ; base d'images ; indexation par le contenu ; navigation psycho-visuelle ; analyse multir{\'e}solution ; classification ; organisation hi{\'e}rarchique ; arbre de recherche flou.},
  TYPE = {Theses},
  PDF = {https://theses.hal.science/tel-00079897v1/file/jlandre_these.pdf},
  HAL_ID = {tel-00079897},
  HAL_VERSION = {v1},
}
```

mybiblio.bib × article-05.tex

mybiblio.bib > ...

```bibtex
@ARTICLE{10891909,
  author={Diaw, Moustapha and Retraint, Florent and Morain-Nicolier, Frédéric and Delahaies, Agnès and Landré, Jérôme},
  journal={IEEE Transactions on Signal Processing},
  title={New Statistic Detector for Structural Image Similarity},
  year={2025},
  volume={73},
  number={},
  pages={1168-1183},
  keywords={Feature extraction;Detectors;Convolutional neural networks;Transformers;Image quality;Noise;Computational modeling;Indexes;Image classification;Training;Classification;local dissimilarity map;weibull distribution;GLRT;similarity measures;pre-trained CNN},
  doi={10.1109/TSP.2025.3543207}}

@phdthesis{landre:tel-00079897,
  TITLE = {{Analyse multir{\'e}solution pour la recherche et l'indexation d'images par le contenu dans les bases de donn{\'e}es images - Application {\`a} la base d'images pal{\'e}ontologique Trans'Tyfipal}},
  AUTHOR = {Landre, J{\'e}r{\^o}me},
  URL = {https://theses.hal.science/tel-00079897},
  SCHOOL = {{Universit{\'e} de Bourgogne}},
  YEAR = {2005},
  MONTH = Dec,
  KEYWORDS = {psycho-visual browsing ; fuzzy search tree. ; hierarchical organisation ; multiresolution analysis ; content-based image indexing and retrieval ; images database ; arbre de recherche flou ; base d'images ; indexation par le contenu ; navigation psycho-visuelle ; analyse multir{\'e}solution ; classification ; organisation hi{\'e}rarchique ; arbre de recherche flou.},
  TYPE = {Theses},
  PDF = {https://theses.hal.science/tel-00079897v1/file/jlandre_these.pdf},
  HAL_ID = {tel-00079897},
  HAL_VERSION = {v1},
}
```

```
In the article \cite{10891909}, the authors used \LaTeX{} to write
their research paper.

The Ph.D thesis \cite{landre:tel-00079897} by Jérôme Landré is about
Images classification.
```

PROBLEMS OUTPUT TERMINAL PORTS DEBUG CONSOLE powershell

```
PS C:\Users\lanjer\Desktop\box\ju2025\teaching\courses\semester1\lp2\project-management-latex\latex-code> lualatex ./article-05.tex
/fonts/type1/public/amsfonts/cm/cmr7.pfb><c:/texlive/2024/texmf-dist/fonts/type
1/public/amsfonts/cm/cmsy10.pfb>
Output written on article-05.pdf (5 pages, 611003 bytes).
Transcript written on article-05.log.
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```

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Landre, Jérôme (Dec. 2005). "Analyse multirésolution pour la recherche et l'indexation d'images par le contenu dans les bases de données images - Application à la base d'images paléontologique Trans'Tyfipal". Theses. Université de Bourgogne. URL: https://theses.hal.science/tel-00079897.

TeX article-05.tex > ...

```latex
\documentclass{article}
\usepackage{xcolor} % colors
\usepackage{graphicx} % include images

\usepackage{fontspec} % font selection with LuaLaTeX
\setmainfont{Cantarell} % sans-serif font
%\setmainfont{Cabin} % sans-serif font
%\setmainfont{Georgia} % serif font
%\setmainfont{FreeSerif} % serif font

\usepackage[backend=biber, style=authoryear]{biblatex}
\addbibresource{mybiblio.bib} % Your .bib file

\usepackage{hyperref} % automatic hyperlinks

\title{My first \LaTeX{} document} % title
\author{Jérôme Landré} % author
```

My first LaTeX document

Jérôme Landré

November 5, 2025

Contents

...duction

1.2 Why use LaTeX?

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2 My image

This is an image of a teacher:

You can see in this image a **teacher** in front of a whiteboard.

7 References

Diaw, Moustapha et al. (2025). "New Statistic Detector for Structural Image Similarity". In: IEEE Transactions on Signal Processing 73, pp. 1168–1183. DOI: 10.1109/TSP.2025.3543207.

Landre, Jérôme (Dec. 2005). "Analyse multirésolution pour la recherche et l'indexation d'images par le contenu dans les bases de données images - Application à la base d'images paléontologique Trans'Tyfipal". Theses. Université de Bourgogne. URL: https://theses.hal.science/tel-00079897.

```latex
\usepackage{hyperref} % automatic hyperlinks
\usepackage[left=2cm, right=2cm, top=3cm, bottom=3cm]{geometry}

\title{My first \LaTeX{} document} % title
\author{Jérôme Landré} % author
\date{\today} % date

\begin{document} % body of the document starts here
```

My first LaTeX document

Jérôme Landré

November 5, 2025

Contents

1 Introduction

1.1 Foreword

Let's study a simple derivative in LaTeX:

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The Ph.D thesis Landre 2005 by Jérôme Landré is about Images classification.

2 My image

This is an image of a teacher:

You can see in this image a **teacher** in front of a whiteboard.

You can write with several different sizes using LaTeX:

tiny text
small text
normal text
large text
LARGE text
huge text
HUGE text

3 Using tables

In LaTeX, you can use the tabular environment to create tables:

Teacher	Course	Number of students
Bilal Z.	Research Methods	90
Jérôme L.	Databases	80
Jérôme L.	Operating Systems	50

JÖNKÖPING UNIVERSITY

4 Using figures...

You can also declare a figure, use a caption and a label to reference it later (but you need to compile the file TWICE to get the correct reference).

Figure 1: A lake in Uchon, Burgundy, France.

5 Using lists

5.1 The itemize environment

You can create itemized lists:

- Bilal
- Mira
- Jasmin
- Jérôme
- Maria
- ...

5.2 The enumerate environment

You can create enumerated lists:

1. The winner

2. The second
3. The third
4. The fourth
5. ...

6 Conclusion

In figure 1, you can see a lake in Uchon, Burgundy, France. It is my preferred place to relax when I am home. It is so quiet and peaceful! If you want, you can add the page of the figure like this: see figure 1 on page 3.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

7 References

Diaw, Moustapha et al. (2025). "New Statistic Detector for Structural Image Similarity". In: IEEE Transactions on Signal Processing 73, pp. 1168–1183. DOI: 10.1109/TSP.2025.3543207.

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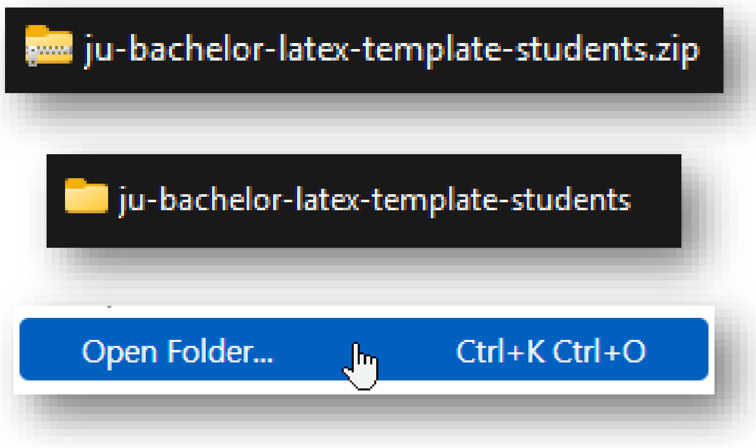
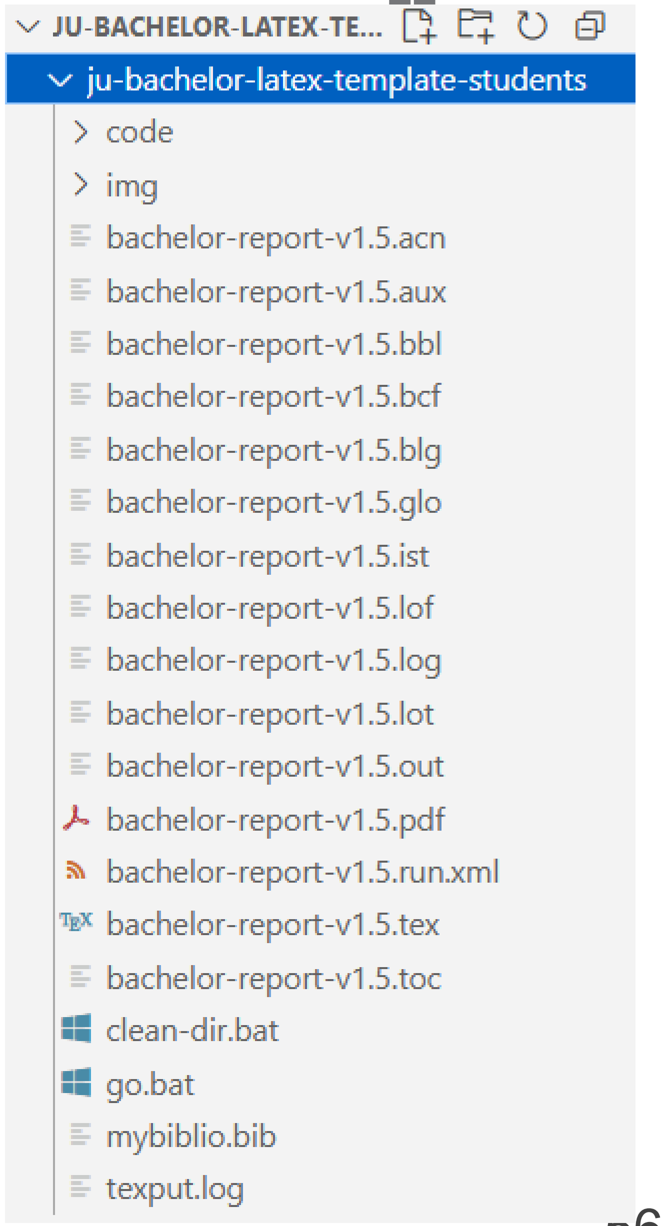
JÖNKÖPING UNIVERSITY

8. Using a template





8. Using a template

bachelor-report-v1.5.tex

ju-bachelor-latex-template-students > bachelor-report-v1.5.tex > ...

```latex
%%%
%%% JU BACHELOR THESIS - LATEX TEMPLATE
%%% Jérôme Landré - Jönköping University - Sweden
%%% Version 1.5 - 2024-12-12
%%%
\documentclass[a4paper, 12pt]{report}

%%%
%%% PACKAGES
%%%
\usepackage[english]{babel} % LANGUAGES
\selectlanguage{english}
\usepackage{csquotes} % LANGUAGES
\usepackage[left=2cm,right=2cm,top=2cm,bottom=2cm]{geometry} % MARGINS

\usepackage{xcolor} % COLORS
% JU color definitions according to the "graphic manual 2023"
\definecolor{jupurple}{RGB}{149, 26, 128}
\definecolor{juyellow}{RGB}{255, 181, 0}
\definecolor{juturquoise}{RGB}{85, 170, 167}
\definecolor{jublue}{RGB}{0, 56, 101}
\definecolor{jugrey}{RGB}{121, 121, 121}

\usepackage{graphicx} % INCLUDE IMAGES

\usepackage{multicol} % MULTICOLUMNS
\usepackage{anyfontsize} % FONT SIZE
\usepackage{hyperref} % HYPERLINKS
\usepackage{listings} % LISTINGS from https://www.overleaf.com/learn/latex/Code_listing

\usepackage{lipsum} % FILLING TEXT LOREM IPSUM
\usepackage{indentfirst} % INDENT THE FIRST LINE OF ANY TEXT

% COLORS FOR THE CODE
\definecolor{codegreen}{rgb}{0,0.6,0}
```

8. Using a template

```
%%%-----
%%%
%%% BEGIN DOCUMENT
%%%
%%%-----
\begin{document}



\pagenumbering{arabic} % NUMBERING THE PAGES
\selectlanguage{english} % NUMBERING THE PAGES

\setlength{\parindent}{1cm}
\setlength{\parskip}{0.2cm plus 1mm minus 1mm}

\thispagestyle{empty} % no page number on the title p
```

```
%%%
%%% TITLE PAGE
%%%
{% JU LOGO
\begin{tikzpicture}[remember picture, overlay]
  \node[xshift=5.4cm, yshift=-3.4cm] (logo) at (current page.north) {\includegraphics[height=1.4cm]{img/JU_B_sv.png}};
\end{tikzpicture}
}

{% TITLE
\vspace{6cm}
\parindent=0cm
\textbf{
\fontsize{36pt}{48pt}\selectfont
A LaTeX template for \\
bachelor thesis work at JU\\
% L1liI
}

  {
    \vspace{1cm}
    \fontsize{24pt}{36pt}\selectfont
    A professional way to present your research
  }
}

{
  \begin{tikzpicture}[remember picture, overlay,scale=1]
    %above
    \node[opacity=0.8, rotate=180, yshift=0.7cm] (rectangle_logo2) at (current page.center) {
      \includegraphics[width=21cm]{img/wave1.png}
    };
    %below
```

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8. Using a template

```
%%%
%%% TABLE OF CONTENTS
%%%
\newpage
\tableofcontents
```

```
%%%
%%% INTRODUCTION
%%%
\chapter{Introduction}

\textcolor{jupurple}{
The chapter provides a background and a clear motivation for the study and the problem area the study is addressing. Further, the
purpose and the research questions are presented. The scope and delimitations of the study are also described. Lastly, the disposition
of the thesis is outlined.
}

\section{Problem Statement}

\textcolor{jupurple}{
The problem statement aims to provide a more detailed description of the components involved in the problem depicted in the section
above. It is important to transform the specific problem, e.g. the specific task from a company, to a broader and more universal or
general problem which may be of relevance for a bigger audience, for example other industries or trade sectors. This is done by
presenting a chain of reasoning in which the specifics of the task are portrayed as a special case of something more generic.
Apart from demonstrating a broader need of the specific task it is also important to relate to the “state of the art” research in this
area and present the scientific relevance/gap why this has scientific relevance. Use references according to the APA system. This is
done appropriately by describing the research front for the area with its in-depth parts. By this reasoning X, standpoints are taken
along the way which will lead to a well-argued and traceable specification and funneling of the problem area, whereby the purpose and
the research question will follow naturally.
Start with a wider perspective and narrow down your descriptions in order to end up in clear and well justified research question(s) and
purpose. The reader shall, based on the argumentative problem statement develop a clear understanding for why this is a relevant study
to do, the so-called knowledge gap in the existing literature.
Elaboration of theoretical dimensions in the thesis is then done in the chapter “Theoretical framework”.
}

\medskip
If you want to use acronyms, please look at the source code. On a Linux \acrfull{os} computer, using the C programming language, we want
to compute the number $\pi$ using the formula:

$$
\frac{\pi^2}{6} = \sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots
$$
```

8. Using a template

```
\medskip
The purpose is to compute the correct sum of terms which means avoiding the race condition problem. Using the APA citation model (\cite
{apaju2024}), you can cite a reference in different ways:

\citeauthor{flanagan2001}, \citeyear{flanagan2001};

\autocite[compare to][3]{flanagan2001},

\parencite{flanagan2001},

\textcite[5]{flanagan2001}.



\bigskip % big vertical space
\textbf{Research Question 1:} Why use LaTeX to write my bachelor thesis?

\medskip % medium vertical space
\textbf{Research Question 2:} Is there a more advanced text processor than LaTeX?
```

```
%%%
%%% BIBLIOGRAPHY ENTRIES
%%%

@online{apaju2024,
    title = {The APA citing sources guide},
    organization = {Jönköping University Library},
    date = {2024},
    url = {https://guides.library.ju.se/apa-en}
}

@inproceedings{flanagan2001,
author = {Cormac Flanagan and Stephen N. Freund},
title = {Detecting race conditions in large programs},
year = {2001},
isbn = {1581134134},
publisher = {Association for Computing Machinery},
address = {New York, NY, USA},
url = {https://doi.org/10.1145/379605.379687},
doi = {10.1145/379605.379687},
booktitle = {Proceedings of the 2001 ACM SIGPLAN-SIGSOFT Workshop on Program Analysis for Software Tools and Engineering},
pages = {90--96},
numpages = {7}
}


@ARTICLE{9928253,
  author={Diaw, Moustapha and Landré, Jérôme and Delahaies, Agnès and Morain-Nicolier, Frédéric and Retraint, Florent},
  journal={IEEE Geoscience and Remote Sensing Letters},
  title={Optical Aerial Images Change Detection Based on a Color Local Dissimilarity Map and k-Means Clustering},
  year={2022},
  volume={19},
  number={},
  pages={1-5},
  keywords={Feature extraction;Optical imaging;Image color analysis;Optical sensors;Mutual information;Gray-scale;Entropy;Disjoint information (DI);k-means clustering;local dissimilarity map (LDM);mutual information (MI);Weibull distribution},
  doi={10.1109/LGRS.2022.3216952}}

@ARTICLE{5464359,
  author={Guo, Yu and Ruan, Su and Landre, Jérôme and Constans, Jean-Marc},
```

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A LaTeX template for bachelor thesis work at JU

A professional way to present your research

Author(s): Jérôme LANDRÉ & Mary POPPINS

Main subject area: Computer Engineering

School: School of Engineering

Date: June 2024

Contents

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1
Introduction

The chapter provides a background and a clear motivation for the study and the problem area the study is addressing. Further, the purpose and the research questions are presented. The scope and delimitations of the study are also described. Lastly, the disposition of the thesis is outlined.

1.1 Problem Statement

The problem statement aims to provide a more detailed description of the components involved in the problem depicted in the section above. It is important to transform the specific problem, e.g. the specific task from a company, to a broader and more universal or general problem which may be of relevance for a bigger audience, for example other industries or trade sectors. This is done by presenting a chain of reasoning in which the specifics of the task are portrayed as a special case of something more generic. Apart from demonstrating a broader need of the specific task it is also important to relate to the "state of the art" research in this area and present the scientific relevance/gap why this has scientific relevance. Use references according to the APA system. This is done appropriately by describing the research front for the area with its in-depth parts. By this reasoning X, standpoints are taken along the way which will lead to a well-argued and traceable specification and funneling of the problem area, whereby the purpose and the research question will follow naturally. Start with a wider perspective and narrow down your descriptions in order to end up in clear and well justified research question(s) and purpose. The reader shall, based on the argumentative problem statement develop a clear understanding for why this is a relevant study to do, the so-called knowledge gap in the existing literature. Elaboration of theoretical dimensions in the thesis is then done in the chapter "Theoretical framework".

If you want to use acronyms, please look at the source code. On a Linux Operating System (OS) computer, using the C programming language, we want to compute the number π using the formula:

$$\frac{\pi^2}{6} = \sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots$$

It is a classic problem of computing but we want to make it parallel for modern multi-core processors. Of course, we will not go to infinity, no time for this...

You can reference figures and tables wherever you want using the label and ref instructions. For instance, see table 4.1 and figure 4.2, it works!

1.2 Purpose and Research Questions

Clearly and concisely state the purpose and the research questions. The purpose defines what will be performed, examined, or compiled as well as its significance. The purpose can be broken down into research questions which you will answer in your report to fulfil the purpose. Your purpose and research questions will set the path in terms of choices of methods, theories, analysis options, ...Drawing on the problem statement, it is evident that XXX. Further, it is evident that XXX. Consequently, the purpose of this study is: To XXX When the reader has arrived at the research questions then they should appear as a natural result of the all the previous text in this chapter, i.e. the reader shall with ease understand that these are natural and well justified research questions. The reader shall NOT have to read between the lines and make interpretations in order to understand the choice of research questions. In order to fulfil the purpose, you have defined X research questions.

Discuss and argue for the first research question and hence, the study's first research questions is:

1. XXX

Discuss and argue for the second research question and hence, the study's second research questions is:

2. XXX

Discuss and argue for the third research question and hence, the study's third research questions is:

3. XXX

The purpose is to compute the correct sum of terms which means avoiding the race condition problem. Using the APA citation model ("The APA citing sources guide", 2024), you can cite a reference in different ways:

Flanagan and Freund, 2001;

(compare to Flanagan & Freund, 2001, p. 3),

(Flanagan & Freund, 2001),

Flanagan and Freund (2001, p. 5).

Research Question 1: Why use LaTeX to write my bachelor thesis?

Research Question 2: Is there a more advanced text processor than LaTeX?

1.3 Scope and Limitations

Here you should fill in the gaps concerning the purpose and research questions by describing what the study entails, and even more importantly, what the study does NOT entail. Keep in mind that a thorough and coherent problem statement section will result in less need for delimitations. Figure X: The scope and delimitations of the study

In this study, Linux will be used as the preferred Operating Systems. No other OS will be used: no Windows, no MacOS. This paragraph gives an example of the use of the glossary, please look at the LaTeX source code.

If we need to have two columns text, we can do it as well (but it is not mandatory). So nice!

3

Theoretical Framework

Based on your review of "state of the art" (current literature) in the problem statement as well as the developed research questions and purpose, you identify relevant sources/theories that further needs to be addressed to achieve stipulated learning outcomes. The framework is for instance used as a base to generate interview guides, survey questions etc. The framework also serves as a base for the analysis where you let your results from the empirical work meet the results from your theoretical work. In this chapter it is suitable to put forward argument that not only describes different theories but also compares and evaluate them depending on how they will be used in the thesis work. It is important that you refer to all sources used in this chapter according to established rules for referencing (The APA system).

We can find information about Biomedical images in (Guo et al., 2010) and about the Local Dissimilarity Map in (Diaw et al., 2022).

The full program to run the computation of π is given below. It is an external file that is read and added to the document.

```
#include <stdio.h>
#include <stdlib.h>
#include <math.h>
#include <pthread.h>

#define ITERATIONS 1000000000

// shared variables between the threads!!!
double sumEven = 0.0;
double sumOdd = 0.0;

// function to compute the Even sum
void *computeSumEven(void *param) {
    double localSum = 0.0;
    for (int i = 1; i < ITERATIONS; i += 2) {
        localSum += 1.0 / (pow((double)i,2));
    }
    sumEven = localSum;
    printf("THREAD EVEN finished\n");
    return NULL;
}

// function to compute the Odd sum
```

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4

Results

A chapter about results is often divided into two sections, 1) presentation of collected data and 2) data analysis. In the first section the collected data should be presented in an objective and coherent way without personal interpretations, views and evaluations. In the second section, the actual data analysis is presented. This means that the works procedure of the data analysis that previously was presented in the method chapter is populated with content. The collected data is put through the analysis process. The result from this data analysis should generate results that answers your research questions and fulfils your purpose.

4.1 Presentation of Collected Data

In the result, we can have a table (see table 4.1) that describes the data:

Number of terms	Time (sec.)	Approximation of π
1 000 000	0.03	3.140043
10 000 000	0.27	3.1411012
100 000 000	2.42	3.141528
1 000 000 000	21.07	3.141591

Table 4.1: The computation of π for a certain number of terms, the duration in time, and the obtained approximation.

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

4.2 Data Analysis

In LaTeX, you can include images from PNG, JPG, etc. file formats. You can also references the images with label and ref, please see figure 4.2.

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Bibliography

The apa citing sources guide. (2024). Jönköping University Library. https://guides.library.ju.se/apa-en

Diaw, M., Landré, J., Delahaies, A., Morain-Nicolier, F., & Retraint, F. (2022). Optical aerial images change detection based on a color local dissimilarity map and k-means clustering. *IEEE Geoscience and Remote Sensing Letters*, *19*, 1–5. https://doi.org/10.1109/LGRS.2022.3216952

Flanagan, C., & Freund, S. N. (2001). Detecting race conditions in large programs. *Proceedings of the 2001 ACM SIGPLAN-SIGSOFT Workshop on Program Analysis for Software Tools and Engineering*, 90–96. https://doi.org/10.1145/379605.379687

Guo, Y., Ruan, S., Landre, J., & Constans, J.-M. (2010). A sparse representation method for magnetic resonance spectroscopy quantification. *IEEE Transactions on Biomedical Engineering*, *57*(7), 1620–1627. https://doi.org/10.1109/TBME.2010.2045123

9. PGF/TikZ

- In LaTeX, you can always use images to add to your documents
- But you can also use **PGF/TikZ**:

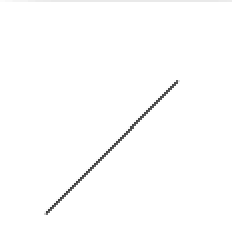
What is PGF/TikZ?

- **PGF** (Portable Graphics Format) is a low-level language for describing vector graphics.
- **TikZ** is a higher-level, user-friendly syntax built on PGF, making it easier to create complex diagrams, flowcharts, plots, and illustrations.
- Both are widely used in academic papers, presentations, and technical documentation for their precision and integration with LaTeX.

Source: Mistral AI, https://chat.mistral.ai/

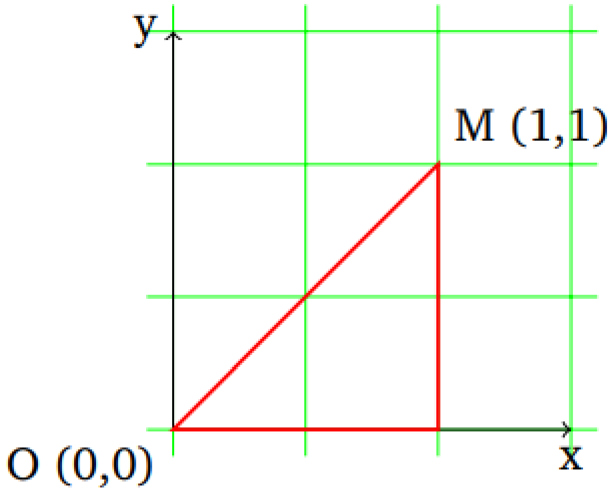
9. PGF/TikZ

```
\documentclass[11pt]{article}

\usepackage{xcolor} % package pour les couleurs
\usepackage{tikz} % package principal TikZ
 \usetikzlibrary{arrows} % librairie optionnelle PGF

\begin{document}
Bla bla bla\dots
\bigskip

\begin{tikzpicture} % figure TikZ
 \draw (0,0) — (1,1);
\end{tikzpicture}

\end{document}
```



Source: https://jlandre.mmi-troyes.fr/docs/pdf/tikz.pdf

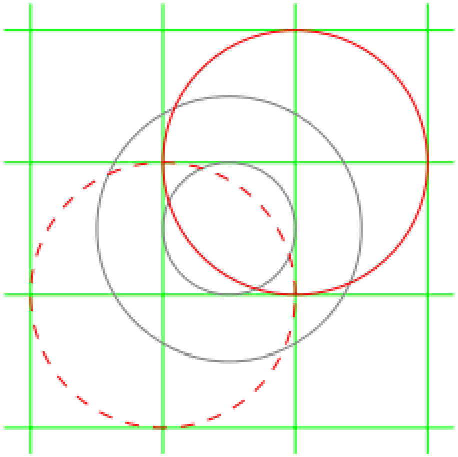
9. PGF/TikZ




```
\begin{tikzpicture}
 \draw [step=1cm,color=green,very thin] (-0.2,-0.2) grid (3.2,3.2);
 \draw [->] (0,0) -- (3,0);
 \draw [->] (0,0) -- (0,3);
 \path (3,0) node [anchor=north] {x};
 \path (0,3) node [anchor=east] {y};
 \draw (0,0) node [anchor=north east] {O (0,0)};
 \draw (2,2) node [anchor=south west] {M (1,1)};
 \draw [color=red,thick] (0,0) -- (2,0) -- (2,2) -- cycle;
\end{tikzpicture}
```

Source: https://jlandre.mmi-troyes.fr/docs/pdf/tikz.pdf

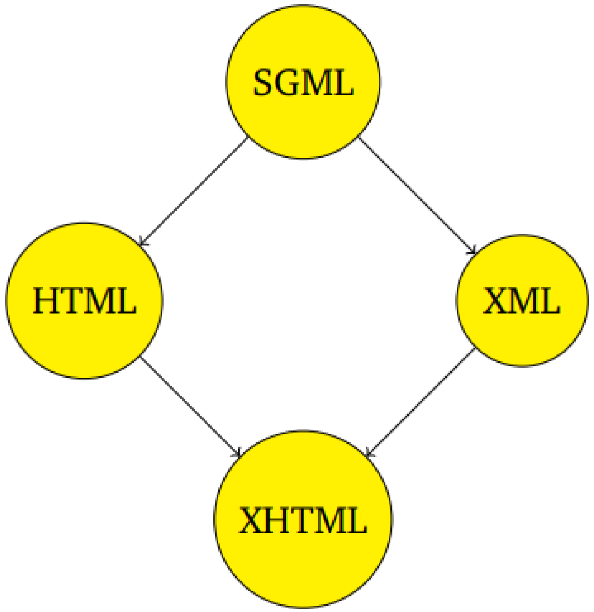
9. PGF/TikZ



```
\begin{tikzpicture}
 \draw[step=1cm, color=green, very thin] (-0.2,-0.2) grid (3.2,3.2);
 \draw [color=gray, very thin]  (1.5,1.5) circle (1cm);
 \draw [color=gray, thin] (1.5,1.5) circle (0.5cm);
 \draw [color=red, dashed] (1,1) circle (1cm);
 \draw [color=red] (2,2) circle (1cm);
\end{tikzpicture}
```

Source: https://jlandre.mmi-troyes.fr/docs/pdf/tikz.pdf

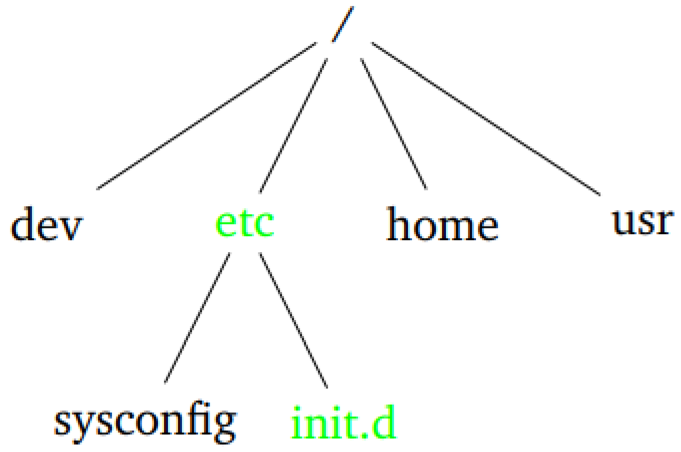
9. PGF/TikZ




```
\begin{tikzpicture}
 \path (0,0) node (n0) [circle , fill=yellow ,draw,inner sep=5pt] {SGML};
 \path (−2,−2) node (n1) [circle , fill=yellow ,draw,inner sep=5pt] {HTML};
 \path (2,−2)  node (n2) [circle , fill=yellow ,draw,inner sep=5pt] {XML};
 \path (0,−4)  node (n3) [circle , fill=yellow ,draw,inner sep=5pt] {XHTML};
 %\pgfsetlinewidth{0.02cm};
 \draw[−>] (n0) — (n1);
 \draw[−>] (n0) — (n2);
 \draw[−>] (n1) — (n3);
 \draw[−>] (n2) — (n3);
\end{tikzpicture}
```

Source: https://jlandre.mmi-troyes.fr/docs/pdf/tikz.pdf

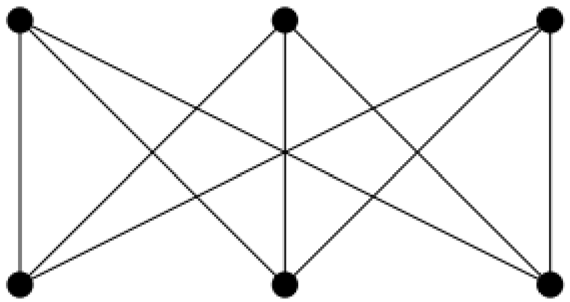
9. PGF/TikZ




```
\begin{tikzpicture}
\node {/}
 child {node {dev}}
 child {node [color=green] {etc}
  child {node {sysconfig}}
  child {node [color=green] {init.d}}
 }
 child {node {home}}
 child {node {usr}};
\end{tikzpicture}
```

Source: https://jlandre.mmi-troyes.fr/docs/pdf/tikz.pdf

9. PGF/TikZ



```
\begin{tikzpicture}
 \foreach \i in {1,...,3}
 { \path (\i,0) coordinate (X\i);
   \fill (X\i) circle (0.5mm); }
 \foreach \j in {1,...,3}
 { \path (\j,1) coordinate (Y\j);
   \fill (Y\j) circle (0.5mm); }
 \foreach \i in {1,...,3}
 { \foreach \j in {1,...,3}
  { \draw (X\i) -- (Y\j); } }
\end{tikzpicture}
```

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To go further…

- LaTeX official documentation:
  - https://www.latex-project.org/help/documentation/
- LaTeX Cheat Sheets:
  - https://quickref.me/latex.html
  - https://guides.lib.chalmers.se/overleaf_latex/tools
  - https://www.bu.edu/math/files/2013/08/LongTeX1.pdf
  - https://www.datacamp.com/cheat-sheet/latex-cheat-sheet
  - https://www.glukhov.org/post/2024/12/latex-cheat-sheet/
- LaTeX examples of use cases:
  - https://www.overleaf.com/gallery/tagged/cheat-sheet
- Bibliography
  - https://www.overleaf.com/learn/latex/Bibliography_management_in_LaTeX

Source: perplexity AI, https://www.perplexity.ai

Thank you for your nice attention,
Do you have questions?